MILITARY SPECIFICATION

TRUCKS, STRADDLE-CARRY DIESEL, 30,000 POUNDS (13,605 kg) CAPACITY, PNEUMATIC RUBBER TIRES, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers heavy duty, diesel powered, pneumatic rubber tires straddle-carry trucks.

1.2 Classification/Specification part number. The straddle-carry trucks shall be of the following sizes as specified herein. Pallet load sizes given are actual dimensions and do not include required load shoe clearances. Height dimensions do include pallet height. Pallets are to be handled by the load shoes without the use of bolsters. The list of required sizes and special shoe lengths (when required) shall be as specified in the ordering data (see 6.2). The classification/specification part number will be formatted to identify each item covered by this specification. The part number will be formatted by selecting from the requirement options available in this specification as follows:

Beneficial comments (recommendations, additions,deletions) and any pertinent data which may be of use in improving this document should be adressed to: Commanding Officer, Navy Ships Parts Control Center, Post Office Box 2020, Attention: Code 0302, Mechanicsburg, Pennsylvania 17055-0788 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

ASMC N/A FSC 3930

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Definitive classification/specification part number M-16101-X-X

M prefix - an item identified by inch-pound (US) units
Specification number
Size code (See 1.2.1)
Class code (See 1.2.2)

Example: M-16101-B-2, a size code B (48 inches (1220mm) wide and 96 inches (2440mm) high) pallet load size, with Class code 2 (24 foot) shoe length.

1.2.1 Size designator. A one-position alpha field used to show the pallet load size (see Table I).

Table I Pallet Load Size

SIZE CODE	DEFINITION
A	48 inches (1220mm) wide and 108 inches (2745mm) high
B	48 inches (1220mm) wide and 96 inches (2440mm) high
C	48 inches (1220mm) wide and 80 inches (2030mm) high
D	48 inches (1220mm) wide and 60 inches (1525mm) high
E	54 inches (1370mm) wide and 108 inches (2745mm) high
F	54 inches (1370mm) wide and 96 inches (2440mm) high
G	54 inches (1370mm) wide and 80 inches (2030mm) high
H	54 inches (1370mm) wide and 60 inches (1525mm) high

1.2.2 Class designator. A one-position numeric field used to show the foot shoe length (See Table II).

Table II Foot Shoe Length

CLASS CODE	DEFINITION
1	8 foot (2438mm) shoe length
2	24 foot (7315mm) shoe length
3	27 foot (8230mm) shoe length
4	Special length (See 6.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

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SPECIFICATIONS

FEDERAL

W-B-131	Batteries, Storage, Vehicular, Ignition, Lighting, and Starting
ZZ-T-381	Tires, Pneumatic, Vehicular
ZZ-I-550	Inner Tube, Pneumatic Tire

MILITARY

MIL-P-116	Preservation; Methods of
MIL-P-514	Plates, Identification, Instruction and Marking, Blank.
MIL-L-2104	Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-M-3971	Meters, Time Totalizing, Non-Hermetically Sealed, Electrical:General Specification for.
MIL-B-11040	Belt, V , Engine Accessory Drive
MIL-M-21861	Manual, Technical, Materials Handling Equipment Preparation, Contents and Approval
MIL-A-46153	Antifreeze, Ethylene, Glycol, Inhibited, Heavy duty, Single Package

STANDARDS

MILITARY

MIL-STD-130	Identification Marking for U.S. Military Property
MIL-STD-162	Material Handling Equipment, Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Process
MIL-STD-209	Slinging and Tiedown Provisions for Lifting
MIL-STD-461	Electromagetic Emission and Susceptibility Requirement for the Control of Electromagnetic Interference
MIL-STD-1474	Noise Limit For Army Material
MS35844	Plug, Machine Thread, Magnetic, (Drain)

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MS49006	Plugs, Pipe, Magnetic-Headless; Aluminum, Iron or Steel

(Copies of Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc. Publication Standards for Safety for Combustion Engine-Powered Industrial Trucks (UL 558)

(Applications for copies should be addressed to: Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, New York; 207 East Ohio Street, Chicago 11, Illinois; or 500 Samsom Street, San Francisco 11, California).

SOCIETY OF AUTOMATIVE ENGINEERS

SAE Handbook

SAE J96	Flashing Warning Lamp for Industrial Equipment
SAE J185	Access Systems for Off-road Machines
SAE J475	Screw Threads
SAE J492	Rivets & Riveting
SAE J551	Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz)
SAE J636	V-belts and Pulleys
SAE J845	360 Degree Emergency Warning Lamp
SAE J917	Marine Push-Pull Control Cables

(Application for copies should be addressed to the Society of Automative Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania 15086).

3. REQUIREMENTS

3.1 Standard product. Trucks covered by this specification shall be complete and shall be constructed and equipped to meet all requirements specified herein. If disassembly of the truck for shipping purposes is required due to truck size the contractor is responsible for on-site reassembly as specified herein (see 3.7.3.6). The end item and major and minor assemblies shall be new and of a standard design or model product used by the end item manufacturer current at the time of bidding except for such minor modification from manufacturer's standard as may be required to conform to these specification requirements. The component parts of the unit need not be the products of the same manufacturer. However, the manufacturer shall be so established in the industry that prompt and continuing service and delivery of replacement repair parts may be assured.

3.1.1 First article (preproduction model). The supplier shall furnish one or more trucks for examination and testing within the time frame specified (see 6.2)to prove prior to starting production that his production methods and choice of design detail will produce trucks that comply with the requirements of

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this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3). A offical report of the tests and inspections will be prepared and submitted to the Government. Report shall include all inspections and tests specified herein. Final report shall bear the government Quality Assurance Represenative's (QAR) seal indicating approval of the entire report. The approved preproduction model may be presented for acceptance as the last unit for delivery under the contract after being rehabilitated to a new appearance. This shall include furnishing of new tires.

3.1.2 Material. Material not definitely specified shall be at least equal to the end item manufacturer's normal commercial product, and shall meet test requirements specified in section 4. "Normal commercial product" shall be interpreted to mean an item furnished by an end item manufacturer normally engaged in commercial production of the item(s) covered by this specification. The Government shall require any bidder, who proposes to furnish other than a "normal commercial product" or Government specified item to submit, prior to and as a condition precedent to the award of contract, detailed engineering drawings of those parts he proposes to furnish in lieu thereof for approval of the contracting officer.

3.1.3 Design. Design of trucks covered by this specification and all components therein, shall insure safe, efficient, and economical operation under normal service conditions. There shall be no evidence of accelerated wear, or failures, or permanent deformations when operated under normal service conditions.

3.1.3.1 Safety. Unless otherwise specified (see 6.2), the truck shall conform to the applicable requirements of Underwriters' Laboratories, Inc., "Standards for Safety" (UL 558, Type D) unless modified herein. The contractor shall submit to the contracting officer or his authorized representive satisfactory evidence that the trucks furnished under this specification meet the requirements of this paragraph. Acceptable evidence of meeting these requirements shall be a UL Label, Factory Mutual certification, or a certified test report from a recognized independent testing laboratory acceptable to the governement, indicating that the trucks conform to Underwriters Laboratories Standard (UL 558, Type D).

3.1.3.2 Maintainability. Provision shall be made for ready adjustment, servicing, or replacement of fan belt, oil filter and components, clutch, starter, alternator, alternator regulator, battery, wearing parts of steering assembly, tires, wheels, lights, and horn. If hand access openings are used on the exterior of the truck, the edge of the opening shall be smooth and shall be provided with a removable hinged cover, whenever it is required. There shall be no interference to the servicing or draining of oils and lubricants to or from any assembly or components by frame members or other obstructions, and without requiring the use of flexible connection or other special devices. Drain plugs shall provide for complete drainage. All special hand tools required for maintenance of the truck shall be provided and shipped with each truck procured.

3.1.3.2.1 Filling, draining and checking provisions. The hydraulic system reservoir, transmission housing, and engine shall be equipped with a dipstick to determine the fluid level. A sightglass in lieu of dipstick is acceptable for the hydraulic system. The dipstick shall be graduated to show the amount required for filling. Dipsticks shall be accessible without the use of handtools. Time to remove, read and replace each dipstick or read sightglass

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shall not exceed two minutes. Oil filler caps shall be painted commerical red. Other enclosures such as gear cases which contain lubricant shall be equipped with dipsticks or with check plugs of not less than 1/2-inch pipe size to determine the fluid level. Each enclosure shall be equipped for filling and for draining. The drain outlet shall be fitted with a magnetic drain plug having either tapered pipe threads or shoulder and gasket, size conforming to MS35844 or MS49006. In lieu of a magnetic drain plug, a separate magnetic device shall be fastened inside the enclosure at or near the bottom and shall be removable for cleaning without removal of other parts or components. Each drain plug or valve shall be located so that removal of the plug or valve will result in complete drainage of the fluid from the enclosure (except torque converter). When the truck is in a level position, drainage of fluids shall be to the ground without draining on or over any part of the truck. Integral tubes, hose, or troughs may be used to convey the fluids from the drain outlet. Accessibility to the fill opening, the fluid-level checking device, and the drain plug or valve shall be provided without removal or adjustment of accesories and parts, except plates equipped with hand-operable, quick-disconnect fasteners, or capscrews in applications where quick-disconnect fasteners are subject to damage. The drain outlets for the engine-cooling system shall provide complete drainage of the system. The radiator drain shall be such that coolant will not drain onto any part of the truck. Extension hose is permitted.

3.1.4 Certification of compliance. When specified herein, (see 6.2) or upon request, a certificate of compliance with the requirements of this specification shall be furnished. Such certificates shall be supported by back-up data which shall be available for review. The certificate of compliance shall read as follows:

> This certifies that the .....(component, part or assembly) complies with all the provisions and requirements of Specification....., paragraph.....

The certificate shall be signed by a duly authorized agent of the contractor.

3.1.5 Operating temperature. Unless specified otherwise (see 6.2), truck operation and engine shall start without preheating within 5 minutes and the truck shall operate as specified herein within 15 minutes after engine start in any ambient temperature from zero °F (-18°C) to 110°F (44°C) when tested in accordance with test method no. 12.

3.2 PROPULSION

3.2.1 Engine. The basic engine shall be a commercial industrial diesel type and as described hereinafter. All other requirements including electromagnetic interference suppression, cooling system, fuel system, starting system and accessories shall be as specified herein. The engine shall be equivalent in design, power and size to the engine normally furnished by the end item manufacturer in similar commercial products and shall provide sufficient power and torque to meet the performance and operational requirements specified herein. Engine accessories shall comply with requirements of 3.2.2.

3.2.2 Engine accessories.

3.2.2.1 Alternator. A 12 volt alternator shall be provided. It shall have output characteristics matched to the electrical load required by the equipment as furnished plus the additional reserve capacity to support the complete

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electrical system and accessories furnished with the vehicle and a 25 percent safety factor capacity. The alternator shall not utilize externally mounted selenium rectifiers. The alternator shall be driven by single or multiple V-belts conforming to MIL-B-11040 to provide full alternator output at mid-range engine speeds without belt slippage. A protective device shall be provided and matched electrically to the alternator so that the vehicle's electrical system including alternator is protected from damage when incorrect polarity is applied or arc welding maintenance repairs are performed anywhere on the equipment. Fuses are not acceptable.

3.2.2.2 Alternator regulator. An alternator regulator shall be supplied if required by the alternator and electrical system design. Provision shall be made within the electrical system for:

(a) Alternator voltage regulation.
(b) Current limiting to protect alternator enclosed diodes or the alternator if applicable.
(c) Disconnection of circuit to the alternator rotor when the ignition switch is off.
(d) Actuation of ammeter or charge indicator lamps where applicable.
(e) Field current relay switch to prevent rotor current from passing through the ignition switch where applicable.

3.2.2.3 Starting system. The starting system shall operate on 12 volt and shall include a starting motor solenoid, a starting motor, necessary wiring and cable and starting motor switch. Cranking torque of the starting motor shall be sufficient to start the engine with battery specified for the truck when both engine and battery are thoroughly cold soaked, at zero °F (-18°C) unless specified otherwise (see 6.2). When specified (see 6.2),a certificate of compliance is required. The starting system shalll be fitted with a device which shall allow starter engagement when the transmission shift lever is in the neutral position only and shall also be protected by a system which shall automatically de-energize the starter when the engine starts and shall prevent starter engagement while the engine is running.

3.2.2.4 Fuel system. The capacity of the fuel tank shall insure not less than 8 hours of continuous operation as measured during Test Method No. 14. The tank shall be equipped with a safety filler cap assembly which shall incorporate a self-closing cap, screw-locking device, removable strainer and provide for padlocking. The cap assembly and the neck of the fuel tank shall be threaded and when installed, the cap shall not be removable without the use of tools. The fuel tank shall be equipped with a stand pipe or other suitable means at the fuel line inlet to prevent foreign matter from entering the fuel line and to prevent drawing fuel from bottom of tank. The fuel tank shall be equipped with a magnetic drain plug at the lowest point of the tank. The drain plug shall be removable with hand tools without the removal of any other component and protected by position or otherwise from external damage. A shut-off valve shall be located immediately adjacent to the full fuel tank. Adequately sized fuel lines shall be provided to insure continuous operation at full throttle. Other construction and design details shall be in accordance with Underwriters' Laboratories, Inc., Publication UL 558. No provisions for auxiliary fuel supply are required. A water separator, fuel filter and metal sediment bowl of adequate capacity shall be provided.

3.2.2.4.1 Air Cleaner. The air cleaner shall be either the dry type or the oil bath automotive type and shall have a capacity sufficient to allow the

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engine to operate under all conditions specified herein without drawing oil from the air cleaner at full throttle. The air cleaner shall be constructed so as to permit disassembly without tools for cleaning or renewal of filter element. It shall be bracket or throat mounted, connected to the air inlet, and accessible when engine cowl, hood or access plate is opened. The air cleaner shall be located or designed to prevent water entering the cleaner when the vehicle is operated during inclement weather.

3.2.2.5 Cooling System. The cooling system shall be the closed pressure type and shall be adequate to prevent coolant temperatures at the head outlet in excess of 115 °F (46°C). over the maximum ambient temperature as specified herein or when the truck is operated continuously with rated loads over the test course. One drain cock shall be located in the base of the radiator or pump inlet elbow, whichever is lower, and the second shall be located at the lowest point of the liquid coolant jacket in the cylinder block. Unless otherwise specified (see 6.2), the truck shall be shipped with engine coolant.

3.2.2.5.1 Fan. The fan shall be a propulsion or suction type rotating on anti-friction bearings and driven by a "V" belt conforming to SAE J636. The belt shall be capable of being replaced and adjusted without requiring more than 30 minutes for the change and without removing any component other than the power take-off universal joint. Fan belt(s) take-up shall be at least three (3) percent of the belt circumferential length, plus an allowance for replacement of a new belt(s) without forcing. The belt shall be clear of all other components within the adjustment range. Polyflex fan belts are acceptable.

3.2.2.5.2 Water Pump. The water pump(s) shall be of the centrifugal type, shall be replaceable, and shall be equipped with shaft seals that will prevent leakage. The shaft shall be corrosion resistant when exposed to water. A lifetime lubrication system shall be required or an adequate and readily accessible pressure lubrication fitting shall be provided.

3.2.2.5.3 Thermostat. A nominal 170 to 195°F (76 to 91°C) thermostat shall be located in the cylinder head outlet to the radiator. It shall be easily removable. Thermostat located in radiator hose is not acceptable.

3.2.2.5.4 Radiator. The radiator assembly shall consist of a fin and tube or honeycomb type core with an expansion tank and sediment tank. Where the top of the radiator is lower than the top outlet of the engine, a surge tank may be used. The radiator cap shall be accessible without physical removal of the engine hood. The radiator shall be mounted to prevent its being damaged because of shock and racking experienced in normal operation of the equipment. The radiator shall be protected by a heavy grille. To prevent abrasion, all hose shall have adequate clearance of the truck structure or other components under all operating conditions.

3.2.2.5.5 Gaskets, Seals and Packing. All gaskets, seals and packing used in the cooling system shall be made of materials capable of resisting deterioration from mixtures of water and ethylene-glycol-base anti-freeze, conforming to MIL-A-46153.

3.2.2.6 Governor. The governor shall be adjusted to limit the maximum speed of the engine to not in excess of the engine manufacturers' published recommended maximum governed engine speed for industrial trucks application. The governed speed of the engine shall limit the maximum forward speed of the vehicle as required in 3.6.5.

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3.2.2.7 Exhaust System. The exhaust system shall conform to structural and safety requirements of Underwriters' Laboratories, Inc., "Standards for Safety (Power-Operated Industrial Trucks)" (UL 558), Type D. The tailpipe shall not extend beyond the plan outline of the body of the truck. A vertical exhaust discharge extending to the roof of the enclosed cab with rain cap is required. The exhaust outlet shall be arranged to prevent burns to the operator during normal operations. The muffler shall be independently supported around the body or muffler outlet. If the tailpipe is independent of the muffler, or extends more than 12 inches from the muffler, it shall be supported.

3.2.2.7.1 Spark Arresting Device. A backfire deflector, when required, will be specified (see 6.2).Device shall conform to UL 558, Type DS.

3.2.2.8 Cold Weather Starting System. Integral starting aids including glow plugs, fluid or oil heaters, fluid priming or battery blankets shall be provided to enable engine to be started under operating temperatures specified herein. If cold weather starting aids uses ether, then a measured shot system shall be employed that is not functional before ignition switch is energized to crank engine and is not functional after engine has started.

3.2.3 Drive Assembly. The drive assembly shall consist of all components necessary to transmit power from the engine to the wheels. Each gear, shaft, and axle shall be of hear-treated alloy steel. All rotating shafts and axles shall be supported on anti-friction bearings. All gears shall operate in lubricant, and a standard pressure grease lubrication system shall be provided for all friction parts not so lubricated. Gear housing shall be equipped with filler plugs and magnetic drain plugs. Differential ring and pinion gears, when used, shall be adjustable. All levers to gear enclosures shall be provided with seals, caps, rubber boots or similar means to prevent the entry of water into the enclosure under normal operation or outside storage. When a drive shaft is provided there shall be a slip joint integrated into the drive line to facilitate the removal of the drive shaft. With drive wheels locked, the drive assembly shall be capable of withstanding 125 percent of the bare engine torque as published by engine manufacturer, multiplied by the stall torque ratio of the torque converter applied to the input shaft of the transmission or a torque computed in foot-pounds (ft-lbs) by the formula

$$T = \frac{.04 \text{ Wheel Dia. (in.) X Gross Veh. Wgt., where .04 is the}}{\text{Gear Ratio}}$$

conversion factor to express the torque in ft-lbs. Whichever is the greater torque shall be applied. Each component shall withstand the test torque without failure, crack, or permanent deformation. The hydraulic pressure may be increased or the clutch locked by external means to prevent slippage at the discretion of the truck manufacturer. In all cases, the transmission shall be in lowest gear range. When test is required (see 6.2), the drive assembly may be tested before or after truck is completely assembled.

3.2.3.1 Transmission. Unless otherwise specified a mechanical transmission shall be furnished (see 6.2). Trucks(s) shall have not less than five (5) speed ranges foward and reverse.

3.2.3.1.1 Power shifted transmission. When specified (see 6.2),the power shifted transmission shall be of the continuous drive type within each speed range(s). An oil filter is required for the transmission oil. Forward and reverse selection lever(s) shall be easy to operate and positive in action without using a manually operated mechanical clutch. The transmission shall be equipped with a removable dipstick marked with the correct fluid level. Truck(s) shall have not less than three (3) speed ranges forward and reverse.

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3.2.3.1.2 Mechanical Transmission. The mechanical transmission shall utilize a hand operated lever for gear shifting between speed ranges and foot operated mechanical clutch to disengage the engine from the transmission. The clutch shall be oil type and of the proper design and characteristics to withstand stresses induced by the engine and transmission and shall provide smooth acceptance of power when clutch is operated. Clutch shall be disengaged by depressing a hobbed or rubber capped pedal. Force required to depress the clutch pedal and disengage clutch shall not exceed 50 pounds (23 kg). An easily accessible lubrication fitting for the clutch throwout bearing and clutch shaft shall be provided, so that the bearing may be lubricated without removal of floor plates or other related components. If self-lubricating bearings are used, a plug shall be used in place of the fitting to provide for lubrication of non-self lubricating replacement bearings.

3.2.3.2 Torque Converter. A torque converter shall be supplied with the power shifted transmission, either as a separate converter unit or as an integral package including both torque converter and power shifted transmission. The converter shall be of size and rated capacity to suit the engine supplied.

3.2.3.3 Transmission Controls. The transmission control levers shall be mounted on the right side of the steering column or in a location convenient for right hand operation. The direction shift lever shall be the upper and the speed shift lever shall be the lower if both are mounted on the steering column. The direction shift lever shall pass through neutral while being moved from forward to reverse. An interlock shall allow starter operation only when the direction shift lever is in neutral. A data plate(s) shall be provided; decalcomania are not acceptable.

3.2.3.4 Drive Unit.

3.2.3.4.1 Drive Shaft. If transmission and differential are not integral, a double universal joint type drive shaft, conforming to accepted automotive practice, shall link the transmission to the differential.

3.2.3.4.2 Differential. The differential shall be of the heavy duty automotive type. The pinion shall be mounted on roller bearings. The ring gear and gear assembly, if used, shall be mounted on roller bearings. The differential shall be mounted as a unit in a removable carrier at the rear of the truck. All gears shall operate in lubricant. When specified (see (6.2),a certificate of compliance is required.

3.2.3.4.3 Drive Axles. The drive axles with universal joints shall link the differential to the driving heads at the rear corners of the vehicle.

3.2.3.4.4 Drive Heads and Chains. Drive heads shall be mounted on the top rear of the vehicle frame over the wheel fork tubes. Drive sprockets and shafts shall be supported in the housings by anti-friction bearings. Automatic drive chain adjustment of not less than four links shall be provided. Chains shall be strong enough to withstand the drive assembly torque test when specified (see 6.2) and the test specified herein without failure, crack, or permanent deformation.

3.2.4 Hoist Mechanism. The hoist mechanism shall be either a mechanical power take-off or a hydraulic type. The hoist mechanism shall show no permanent deformation or failure following application of a 200 percent overload test as indicated in Test Method No. 8.

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3.2.4.1 Mechanical System. If furnished, power used for the hoist and shoe swing mechanism shall be furnished by a mechanical power take-off.

3.2.4.1.1 Power Take-Off. The power take-off shall be coupled to the front of the engine crank shaft by a universal joint or a positive drive conforming to accepted automotive practice. All gear assemblies shall be enclosed in oil tight housings. All gears shall operate in a constant oil bath, and means for checking, draining, and replacing lubricant shall be provided.

3.2.4.1.2 Take-Off Shafts. The take-off shafts shall transmit power from the power take-off to the hoisting and lowering gear boxes. Engagement of the power take-off through a clutching mechanism, shall release the brake which holds the load in position.

3.2.4.1.3 Hoist Brake. Lifting shoes shall be held in position by the hoist brake when the hoist control lever is released or returned to neutral position. Rearward or upward motion of the hoist control lever shall release the hoist brake and raise the load hooks.

3.2.4.1.4 Hoist and Lower Gear Boxes. Hoist and lower gear boxes shall be totally enclosed gear reduction units arranged to transmit power from take-off shafts to side shafts. All gears shall operate in a constant oil bath and means for checking, draining and replacing lubricant shall be provided.

3.2.4.1.5 Side Shafts. Side shafts shall be provided to transmit power from hoist and lower gear boxes to the gears or roller chain sprockets for actuating the shoe lift rods. Side shafts shall be supported on antifriction bearings.

3.2.4.1.6 Shoe Lift Rods. Shoe lift rods or shoe guides shall be provided which are pivoted or otherwise arranged to permit lift shoes to elevate and lower, as well as swing in and out. Power shall be transmitted from side shafts to shoe lift rods or shoe guides by means of rack and pinion gearing or a chain and sprocket arrangement. The lifting mechanism shall be actuated by a hand lever located convenient to the operator. A second lever or automatic device shall be provided to actuate the transverse mechanism. Controls shall be easily operated by the right hand.

3.2.4.1.7 Limit Stop. An automatic limit stop shall be provided which shall halt movement of the hoist mechanism automatically at the limit of travel in either direction, disengage the hoist, and return the hoist lever to a neutral position. The lower limit of travel of the hoist mechanism shall be easily adjusted for varying ground clearance.

3.2.4.1.8 Load Shoes. Load shoes shall extend longitudinally between the wheels along each side of the truck. They shall be made of heavy steel and shall be connected to the elevating mechanism. Load shoes shall be not less than 8 feet long and of sufficient width to support the load securely.

3.2.4.2 Hydraulic System. When a hydraulic pump is furnished to power the hoist and shoe swing mechanism, it shall be direct or chain driven by the engine. The hoist system shall providefor safe lowering of load in event of failure or damage to hydraulic lines. The hydraulic sump tank and system shall be capable of being drained without spillage. The system shall be provided with a pressure relief valve. The system shall have a maximum working pressure (relief valve setting) not in excess of 2,000 psi (13,789,514 Pa). Oil for the system shall conform to MIL-L-2104, grade 10. If an operating temperature below 0 °F is required (see 6.2) Mil-H-5606 shall be used. All hydraulic component

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materials shall be compatible with these oils and shall operate herein with an oil operating temperature range of zero °F to 225°F (-18 to 107°C). The temperature rise in any portion of the hydraulic system shall not exceed 220°F(104°C) during all operations specified herein. All system components shall withstand a proof pressure that is equal to or greater than 2 times the mximum working pressure (2 times relief valve setting) without external leakage, damage, or permanent deformation. All hydraulic system components should have a minimum burst pressure that is equal to or greater than 3 times the system relief valve setting. The hydraulic system shall be cleaned free of foreign matter (including weld slag and spatter). When a single pump provides the system flow, the hydraulic control valves shall be in series but may be in any order. When a multiple element pump is used, each element shall have all related valves in series but may be in any order. Hydraulic connection ports of the hydraulic system components shall be sealed by the component manufacturer and this seal shall be maintained until final assembly on the truck. A 1/4-inch pipe tap test point shall be located at the outlet of each pump or pressure inlet of each control valve for measuring pressure. Hydraulic components shall provide protection against contamination of hydraulic fluid by direct impingement or seepage of other fluids.

3.2.4.2.1 Hydraulic Cylinders. The hoist system shall be actuated by not less than 2 fully equalized, double acting, hydraulic cylinders, so that lift shoes remain parallel to the axis of the truck throughout their range. All hydraulic cylinders shall be pin, socket, or gimbal mounted, and shall be either fabricated from seamless steel tubing, or welded and redrawn tubing. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final machine honing or rolling of the cylinder. Cylinders shall be so located to provide for ease of maintenance, and replacement. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall have an exclusion device to prevent foreign material and fluids from entering and damaging the seal and other cylinder components. All cylinders shall perform as specified herein without buckling or bending. The bearing surface shall be a material that will not scar, score, or gall the inside surface of the cylinder. The rod shall be hard chrome plated to prevent corrosion.

3.2.4.2.2 Hydraulic Lines. Hydraulic lines shall be of sufficient strength and capacity to withstand capacity loads and tests specified herein. Hydraulic lines shall consist of tubing, hose and fittings. Hose or a combination of hose and tubing shall be used between components which are not mounted to a nonflexing, rigidly connected structural member or rigidly connected subassembly. All lines shall be routed to provide the minimum number of bends, ease of maintenance, and maximum protection; provision shall be made to prevent damage of lines due to chafing. The hose installation, such as angle of fittings and location, shall cause no stress concentration on the hose at the fitting, and the hose shall be supported, when necessary, to eliminate sagging at the fitting. All lines, other than suction lines, return lines and cylinder bleeder lines, shall be considered pressure lines. All lines shall be cleaned free of foreign matter including weld slag and spatter.

3.2.4.2.3 Hydraulic Valves. The hoist and shoe swing mechanisms shall be furnished with separate, double acting, self-centering control valves. These valves shall be equipped with adjustable pressure relief valves. The hydraulic hoist circuit shall be provided with flow control to permit smooth lowering action at constant rate regardless of load and check valves to prevent dropping of load in the event of pressure loss.

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3.2.4.2.4 Reservoir. A reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall include a filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be rigidly mounted to prevent vibration that could result in leaks, loosening of fasteners, brinelling of material, or cracks. The location shall not impair operational visibility or access to either side of the operator's compartment. The oil capacity of the reservoir shall be not less than 110 percent of the fluid required for the hydraulic system operation. The reservoir shall have sufficient capacity to prevent air entering the system with all hydraulic pistons fully extended and sufficient free air capacity to prevent oil being discharged through the reservoir air vent when maximum return flow oil is surged into the reservoir from the system. The 'Add' and 'Full' marks on the dipstick shall indicate the correct oil level when the truck is on level ground, and the forks are at ground level. The filler cap and tube shall be located to provide for filling from a standard 5-gallon (.022 m_3) container and to prevent entrance of contaminants throughout filling and checking operations. The filler tube shall have a removable 30 mesh or finer strainer. The reservoir pump suction openings should be located a minimum of 1 inch (25 mm) above the bottom of the reservoir to prevent entrance of settled contaminants. All surfaces inside the reservoir should be treated to prevent corrosion. Except for sealed reservoirs a breather shall be provided and shall filter incoming air. The breather shall preclude the entrance of water induced by rain. The reservoir should be baffled when necessary to provide a tortuous flow path from the reservoir inlet to the reservoir outlet in order to prevent a direct current between the two ports. All return line flow should enter the reservoir at a point which is below the reservoir minimum 'add' oil level to reduce aeration. A removable strainer of 50 or less mesh openings or a 25 micron or less removable filter shall be installed in either the hydraulic reservoir or the suction line to protect the pump. When a strainer is provided for pump protection, a removable filter must also be provided in the hydraulic system. The reservoir shall have one or more accesses for manual cleaning of the entire inside or be removable for cleaning. The hydraulic level indicating gauge shall be provided within easy access to permit checking hydraulic oil level.

3.2.4.2.5 Lever Arms. Wearing surfaces of lever arms that are used for hoisting, lowering and swinging and shoe lift rods transversely, shall be equipped with bushings and a standard pressure lubrication system.

3.3 STRUCTURE

3.3.1 Frame. The frame and related structure shall be fabricated from structural steel or steel castings at least equal to the size and design of manufactuer's normal commercial frame members. Frame members, bearings and all their joints shall provide a rigid unit structure. Chassis and frame shall be capable of withstanding without permanent distortion, stresses induced by a load and tests equal to 200 percent of rated capacity in the transport position.

3.3.2 Wheels. The truck shall be mounted on individually suspended wheels. Each wheel shall be mounted on an alloy steel spindle and tapered roller bearings. All wheels shall track true in accordance with automotive practice, as determined by their respective radii. There shall be 2 single front steer wheels and 2 single rear steer-drive wheels. The rear wheel hubs shall provide for mounting removable drive sprockets. Wheels shall have demountable type rims which can be removed from the hubs without deflating tires. Each wheel shalll be mounted on wheel studs or capscrews with self-centering tapered ends or on tapered boltholes which shall provide a positive method of locating the wheel on the hub.

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3.3.2.1 Tires. Tires shall be pneumatic, high pressure, non-directional type. The size and ply rating shall be not less than 11.00 x 20 (14 ply). They shall comply with the requirements of Specification ZZ-T 381.

3.3.2.2 Tubes. Tubes shall be heavy duty type and the size required to fit the tires furnished. Valve stems shall be arranged to permit inflation of mounted tires from the side of the truck. The tubes shall comply with the requirements of Federal Specification ZZ-I-550.

3.3.3 Steering. An automotive type steering wheel, mounted on the steerig column shall turn the front and rear wheels. Clockwise rotation of the steering wheel shall provide for right hand turning of the truck while in forward motion. A hydraulic steering booster or other power steering shall be provided. Provision shall be made for precision adjustment to compensate for wear of all major steering parts, and for power steering fluid reservoir. When provided, the reservoir shall be accessible for filling or checking of lubricant level. An unbalance tangential force of not less than 150 pounds (68 kg) applied to the outer periphery and in a plane with the steering wheel shall not permanently distort, fracture or deform any part of the steering mechanism, including the steering column or wheel.

3.3.3.1 Wheel Mounting. Wheel mountings and axles shall be arranged to permit the truck to negotiate the obstacle course in Test Method 14, with no more than one displaced wheel off the ground at any time. Means shall be provided to prevent wheels or tires from contacting any part of the truck structure, irrespective of obstacle magnitude. Axle assemblies, exclusive of axle supports shall be high grade hardened steel. The axle supports shall be of high grade steel.

3.3.3.2 Wheel Forks and Tubes. Wheel forks shall be of steel and designed to permit required wheel angularity, steering radii, and tire accessibility. Fork tubes shall be of sufficient strength and rigidity to withstand, without permanent deformation, stresses induced by a load equal to 200 percent of rated capacity in the transport position. Forks shall be readily detachable from the fork tubes. The minimum interchangeability of forks and tubes shall be from side to side in the front and from side to side in the rear. Wheel forks interchangeable right front to left rear and left front to right rear is also acceptable.

3.3.3.3 Springs. The truck shall be mounted on coil springs providing individual suspension for each wheel. Springs shall be fully enclosed in fork tubes or in separate housings.

3.3.4 Brakes.

3.3.4.1 Service Brakes. Service brakes shall be of the hydraulic, internal expanding or disc type ,adjustable, hydraulically actuated mounted on all wheels. Brakes shall be applied by depressing hob or rubber capped pedal with the right foot. Dual master cylinder shall be provided. Use of a 2-stage master cylinder or power booster is acceptable. When the hydraulic master cylinder is located under the floorboard an access hole of not less than 2-1/2 inches (64 mm) in diameter shall be provided above the master cylinder unless a remote reservior is provided with provisions for filling master cylinder. Time to cneck fluid level in remote reservior shall not exceed two minutes.. The access hole shall be provided with a cover which can be easily removed without the use of any tool. Vacuum power booster, if provided, shall be provided with

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a check valve and a vacuum reservoir capacity which shall provide 3 complete stops from rated speed, when carrying rated load with the engine stopped. Reservoir vacuum shall not decrease more than 70 percent during this test. Brakes shall be located so that hydraulic lines may be bled of air without the necessity of removing the wheels or disconnecting hydraulic lines. Brake lining, may be either molded, woven and bonded or riveted.

3.3.4.2 Mechanical Parking Brake. The parking brake shall be an independent mechanical friction brake mounted on the drive or propeller shaft rotating with the wheels, or a separate mechanical linkage actuating the service brake. It shall be operated by a hand lever equipped with a locking device. The hand lever shall be located within easy reach of the operator and in a position permitting easy and safe movement on or off the vehicle from either side. The parking brake shall be capable of retaining the truck with rated load on a 10 percent grade. Adjustment of parking brake shoes or bands shall be accomplished without the removal of any major assembly other than tires and wheels.

3.3.5 Body. The truck body shall be constructed of heavy steel sheet or plate equal to that used in the manufacturer's normal commercial product.

3.3.5.1 Seat. A cushion seat, having a minimum longitudinal adjustment of 4 inches, shall provide comfortable seating for the operator when operating the truck. The seat shall be provided with a cushioned backrest.

3.3.5.2 Closed Cab. Unless specified otherwise (see 6.2), a closed, weathertight metal cab, designed for maximum visibility, shall be provided. Visibility shall be such as to permit the operator an unobstructed view to the level road in all directions. Unless otherwise specified (see 6.2), cab shall be equipped with one door, two ventilation fans, windshield, windows (doors and rear), windshield wipers (including rear window), heater, and windshield defroster. The cab shall have easy removeable floor boards and wall coverings in the interior of cab to limit heat from the engine and drive train from entering the cab. The floor boards and wall coverings shall be removeable within 10 minutes to permit maintenance. All penetrations of the cab shall also be sealed by rubber boots or covers. Sufficient opening(s) shall be provided in the cab floor to provide load visability, to efficiently straddle the load, and to allow the operator to insure the load is properly engaged with the load shoes. All floor openings required for visability shall be covered with shatter-proof glass.

3.3.5.3 Cowl or Hood. The engine shall be completely enclosed with a quick-opening hood cover or side plates which can be opened without the use of hand tools. Adequate means shall be provided to retain the hood or side openings in an open position to facilitate routine servicing. Louvers or other provisions for ventilation should be provided.

3.3.5.4 Windshield. The windshield assembly shall be a one piece glass unit mounted on the cowl. The windshield shall be of a shatter-proof glass and provide maximum visibility necessary for efficient and safe operations. The windshield shall be equipped with not less than 1 power-operated windshield wiper. The windshield wiper shall be electric or vacuum powered; if vacuum powered, it shall be provided with a booster or vacuum reservoir.

3.3.5.5 Instrument Panel. An instrument panel shall be installed convenient to and plainly visible to the operator. On this panel shall be mounted the

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ammeter, oil pressure gauge, coolant temperature gauge, speedometer, hour meter, fuel gauge, ignition switch, light switch, and starter button or switch specified herein. When a torque converter is supplied, an over temperature indicator for the torque convertor lubricant or fluid shall be mounted on or adjacent to the instrument panel. If an indicator light is used it must be readily visible to the operator. All gages shall be labeled or indicate component and fluid being monitored.

3.3.5.6 Slinging and Tiedown attachments. Vehicles shall be equipped with slinging and tiedown attachments comforming to MIL-STD-209, Class 1, 2 or 3 for Type II or III equipment, excluding air transportation requirements. Attachments shall be indicated on a transportation plate comforming to MIL-P-514 and slinging and tirdown attachments shall identified by stenciling or other suitable markings in appropiate locations on the exterior of the equipment. Slinging and tiedown markings shall indicate clearly the purpose of ech attachment.

3.3.5.7 Hand Holds. Suitable hand holds and steps conforming to SAE J185 shall be provided to assist the operator in mounting the truck from two sides.

3.4 Controls and Instrumentation. Where applicable, the provisions of load handling symbols for powered industrial trucks (ANSI MH11.3), shall apply. Electric controls shall be heavy duty type. Controls and instrumentation shall be constructed to be moisture and weather resistant.

3.4.1 Controls. All switch and lever type controls shall be provided with position marking (decals are not acceptable) either at the control, or a diagram visible to the operator. When a knob is grasped and operated by the operator, his hand shall clear any adjacent knob, lever, fitting, and truck structure by not less than 3/4 inch (19 mm). If mechanical push-pull remote controls are furnished they shall be of the waterproof antifriction type, complying with the requirements for marine cables as specified in SAE J917, except as specified herein. As a minimum, all exposed parts of the controls shall be fabricated of corrosion-resistant steel. The cables shall be covered with a polyethylene or vinyl jacket. Cable ends shall be tailored to actuate all levers through their respective arcs without binding. Bend radii shall be as recommended by the cable manufacturer. The manufacturer's published ratings for continuous operation shall not be exceeded for any required operation of the control specified herein. Forces necessary to actuate direction and hydraulic controls shall be 10 pounds at the handgrip and shall have sufficient strength to withstand without permanent deformation or damage all vibration forces generated during any operation specified herein. The travel of the controls shall not exceed 6 inches from the center position when measured at the end of the grip.

3.4.1.1 Light Switch. An "on-off" light switch shall be located on the instrument panel. When in the "on" position, the light switch shall operate the front headlights and rear tail-lamps, or vice versa, as controlled by the interlocking light switch. The switch shall be protected against adverse weather conditions.

3.4.1.2 Engine Run Control. A key type engine run control shall be furnished. It shall have not fewer than two positions, "on" and "off". The switch shall control the operation of monitors and all electrical equipment except the horn. Three keys shall be furnished with each truck. All trucks under each contract shall be keyed identically. A mechanical engine shutoff may be furnished and shall be in addition to the emergency shutoff as specified.

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3.4.1.3 Starting Motor Switch. The starting motor switch shall be a solenoid plunger type connecting the starting button to the battery. It shall be operated by depressing a button installed on the instrument panel or turning a spring-loaded engine run control to a "start" position. If a push control is used, it shall be inoperable when the engine run control is in the "off" position.

3.4.1.4 Brake Pedal. The brake pedal shall be conveniently located for right foot operation with a clearance of at least 1-1/2 inches (38 mm) on all sides. Brake pedal width shall be at least 2-1/2 inches (64 mm). Suitable means shall be provided for lubricating moving parts of the brake pedal linkage.

3.4.1.5 Accelerator Control. The accelerator control shall be installed for comfortable right foot operation, and shall be located to the right of and shall be free of interference from the brake pedal by a distance of not less than 1-1/2 inches (38 mm). A hand control is not required.

3.4.1.6 Hoist and Shoe Swing Controls. Selective hoist and shoe swing controls shall be self-centering lever(s), except that automatic operation of load shoe swing control shall be permitted. Control(s) shall be conveniently located for right hand or left hand operation. Rearward or upward motion of hoist and load swing control lever(s) shall grapple and raise the load.

3.4.1.7 Load Light Switch. A load light switch shall be provided and located on the instrument panel.

3.4.1.8 Clutch Pedal. The clutch pedal shall be conviently located for left foot operation with a clearance of at least 1-1/2 inches (38 mm) on all sides. Clutch pedal width shall be at least 2-1/2 inches (64 mm). Suitable means shall be provided for lubricating moving parts of the clutch pedal linkage.

3.4.2 Instruments. All gauges and meters shall be products of manufacturers regularly engaged in producing these types of instruments. They shall be flush mounted on the instrument panel. All instruments shall be constructed to be moisture and weather resistant.

3.4.2.1 Ammeter or voltmeter. The truck shall be equipped with a remote indicating ammeter or voltmeter of sufficient capacity to indicate variations in current supply or demand experienced by the truck, exclusive of starter demands.

3.4.2.2 Oil Pressure Gauge. A remote indicating, electrical or bourdon tube type oil pressure gauge shall be installed to indicate true engine lubricating oil pressure.

3.4.2.3 Temperature Gauge. A remote indicating, electrical or capillary tube type coolant temperature gauge shall be provided.

3.4.2.4 Hour Meter. An electric operational hour meter, which registers the number of engine operating hours only, by means of an oil pressure switch, shall be provided. The totalizing mechanism shall register not less than 9999. The meter shall be in accordance with Specification MIL-M-3971, Type I.

3.4.2.5 Speedometer. A combined automotive type speedometer shall be provided. The speedometer shall automatically compensate for forward or rearward motion of the truck. It shall indicate speed of the truck and distance traveled by the truck in either direction.

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3.4.2.6 Fuel Gauge. An electric, thermally stabilized, remote indicating fuel gauge shall be provided.

3.4.2.7 Direction Indicators. Electric automotive type turn indicators shall be mounted on both sides of the front and rear of the straddle truck readily visible from road level in accordance with standard automotive practice to indicate turning direction. A suitable control lever shall be mounted on the steering column convenient to the truck operator. A signal type device readily visible to the operator shall indicate the turn indicator's operation.

3.5 Accessories. All accessories shall be constructed to be moisture and weather resistant.

3.5.1 Headlights. The truck shall be equipped with 2 front and 2 rear headlights, of not less than 25 watts. Lights shall be multiple beam, automotive sealed beam type, with a dimmer switch placed for left foot operation. Directional focus of these lights shall be vertically and horizontally adjustable. Adequate physical protection against damage shall be provided.

3.5.2 Stoplights. The truck shall be equipped with 2 front and 2 rear automotive red reflector type, combination stoplights and tail lamps,of not less than 15 candle power respectively. Stoplights shall operate automatically only upon actuation of the service brake pedal and shall not operate when ignition switch is in an "off" position. Combination stoplights and tail lamps shall be mounted below or adjacent to the headlights and be protected against damage.

3.5.2.1 Wiring and Switching. Wiring and switching shall be such that operation of headlights on one end of the truck shall operate tail lamps on its opposite end.

3.5.2.2 Interlocking Light Switch. A directional switch interlocked with directional gear shift control shall be provided. The switch shall permit the front headlights and rear tail lamps to operate when the vehicle is in neutral and forward gear. When vehicle is in reverse gear, the rear headlights and front tail lamps shall be permitted to operate.

3.5.3 Load Lights. The truck shall be equipped with 2 load lights, of not less than 15 candle power each. Lights shall be mounted on the underside of the vehicle frame, to the left and to the right of the operator, and shall illuminate the load being straddled or lifted.

3.5.4 Battery. Trucks shall be furnished with a 12-volt battery conforming to W-B-131. Wood separators are not acceptable. The battery shall meet the starting requirements of 3.2.2.3. Battery cable ends shall be indelibly marked "POS" or "+", or "NEG" or "-". Unless otherwise specified (see 6.2), the battery shall be charged and dry, and one filling of electrolyte of the required specific gravity shall be shipped with the truck in a separate container.

3.5.4.1 Battery Mounting. Battery mounting shall be under the engine hood or in accessible compartment for easy battery removal. If closed cab is specified (see 6.2), the battery shall not be within the confines of the cab. If mounted outside, the battery shall be housed in an individual acid resistant finisned metal box, incorporating provision for drainage in the lowest points of the battery compartment base. The box shall be fitted with a quick release cover to provide for inspection and servicing. Cover and positioning shall

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protect against short circuiting. The ungrounded cable shall be protected by rubber grommets or insulated passages at entry to battery box. Battery hold-down clamps shall be provided to hold the battery in a fixed position. Battery supports, holddowns, and areas around the installation which could possibly be affected by dripping or seepage of acids shall be protected with acid resistant paints or flexible protective phenolic coatings.

3.5.5 Horn. The truck shall be equipped with 2 electric horns, with push button mounted in the center of the steering wheel. Horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant. A push button type switch shall be provided. One horn shall be mounted near the front and the other near the rear of the truck and shall face to the front and rear of the truck respectively. The two horns shall be interlocked with the direction shift lever so that forward gear provides for operation of the front horn, and reverse gear the rear horn.

3.5.6 Wheel and Drive Safety Guards and Platforms. The truck shall be equipped with suitable full length sheet steel safety guards covering drive chains and sprockets. The truck shall be equipped with suitable sheet steel wheel guards providing protection down to 10 inches (254 mm) above the ground and shall provide sufficient clearance to allow the tires to rotate freely under all operating conditions. Gears, pulleys and all moving parts shall be covered with adequate guards to protect personnel from injury. An open grate safety platform shall be provided on both sides of the engine to provide access to the engine componets. The platform shall conform to SAE J185 and shall extend from the engine to the outside edge of truck. The individual grate openings shall be such that they do not interfere with the operators visibility for loading and unloading operations.

3.5.7 Fire Extinguisher. When specified (see 6.2), each truck shall be equipped with a fire extinguisher. Mounting shall be on a quick release bracket, in an accessible position, with the fire extinguisher nozzle protected against damage.

3.5.8 Dual Rearview Mirrors. The truck(s) shall be equipped with a rearview mirror mounted on the left and right hand sides of the truck within the operators field of view.

3.5.9 Guardrails. The truck(s) shall be equipped with suitable guardrails located on top of straddle truck frame. Guardrails shall encircle the truck perimeter except at ladder locations and they shall conform to SAE J185. Guardrails shall be designed such that they do not interfere with the operator's visability.

3.5.10 Safety Features. Each truck shall be equipped with following safety features:

3.5.10.1 Sounding Device. Directional sounding device, which operates continuously in the mode of travel of the truck. The truck shall have a minimum of two sounding devices, one for forward motion and one for rearward. The sounding device shall be a minimum 100 db and provide an audible warning distinguishable from the normal operational noise of the truck.

3.5.10.2 Lights. Flashing amber lights visible from all directions, which operate continuously when the truck is in operation. The lights shall conform to SAE J96 or SAE J845. A minimum of four lights shall be provided, one on each corner of the truck. The lights shall be mounted between three to five feet

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(915 to 1525 mm) above ground level and shall not interfere with truck operations or exposed to damage from loading and unloading operations.

3.5.10.3 Mirrors. At a minimum (see 3.6.22) a wide angle convex mirror shall be mounted on the right front and left front of each truck. The mirrors shall be a minimum of fourteen (14) inches (356 mm) in diameter and shall provide visibility in front of the truck and along the right side of the truck for left mounted cabs, the opposite shall be provided for right mounted cabs. The mirrors shall be mounted in such a manner as not to interfere with normal truck operations or maintenance procedures. Additional mirrors shall be provided if required to obtain the desired visability based on truck design.

3.6 Performance.

3.6.1 Lifting Speed. The time for grappling and lifting a rated load from the ground to 12 inches (305 mm) above the ground shall be no more than 15 seconds. Lifting speed shall be measured in accordance Test Method No. 1.

3.6.2 Lowering Speed. The time for lowering the unloaded shoes from 10 inches (255 mm) above the ground to ground level and swinging them out to their maximum spread shall not be greater than 15 seconds. The hoist system shall be damped, metered, or easily manually regulated to reduce the shock when a rated load is lowered at maximum speed and approaches ground level. Lowering speed shall be measured in accordance with Test Method No. 1.

3.6.3 Turning Radius. Size 1 through 8 trucks shall be capable of turning right and left in 90° intersecting aisles, 18 feet 0 inches (5485 mm) wide with rated load and a maximum load length of 27 feet (8230 mm).

3.6.4 Steering. The 4 wheels shall be capable of being turned from extreme right to extreme left, or vice versa, in not more than 7 complete turns of the hand steering wheels; and the force required to operate shall not exceed 20 pounds when the truck is stationary, as measured by Test Method No. 2. Other maneuverability features shall meet test course requirements of Test Method No. 14 . The steering wheel free play shall not exceed 3 inches.

3.6.5 Speed. While carrying its rated load on a level surface, each truck shall be capable of a forward speed not less than 25 miles per hour (11.2 m/s) in high gear and a reverse speed not less than 8 and not more than 17 miles per hour (3.6 and not more than 7.6 m/s) in high gear when measured in accordance with Test Method No. 3. If the trucks exceeds 17 mph in reverse a metal warning plate shall be provided in plain view of the operator stating the following: "CAUTION - WHEN IN REVERSE DO NOT OPERATE THIS VEHICLE IN 5TH GEAR OR MORE THAT 17 MPH".

3.6.6 Slope Ascension, First Speed. Trucks shall be capable of ascending a 10 percent grade on a dry paved surface and be able to accelerate from a dead stop on this grade when carrying rated load, as measured by Test Method No. 4.

3.6.7 Underclearance. The truck shall have sufficient underclearance to permit operation from one horizontal plane to another, up or down a 10 percent slope, and at least negotiate the obstacle course described in Test Method No. 14. When carrying rated load at maximum elevation, the truck shall have a minimum underclearance of not less than 7 inches (130 mm).

3.6.8 Service Brakes. A brake pedal pressure between 50 pounds (22 kg) and 150 pounds (68 kg), shall develop a draw bar pull equal to 30 percent of the

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gross weight of the truck (with rated load) while the truck is being towed over a paved surface having a coefficient of friction not less than 0.50. The brake system shall be capable of withstanding brake pedal pressure of 250 pounds (114 kg) without failure of any component. Stopping distance shall not exceed 40 feet (12,192 mm). These requirements shall be measured in accordance with Test Method No. 5, 6 and 7.

3.6.9 Lift Assembly. The lift assembly shall show no appreciable signs of wear after completion of the performance and tests over the test course under the provisions of Test Method No. 14.

3.6.10 Truck Frames, Wheel Forks, Tubes, and Hoist Mechanism. The truck shall support a load of 200 percent rated capacity in the transport position with no permanent deformation or failure. The 200 percent overload shall be applied as indicated in Test Method No. 8. Blocks may be used to relieve load on tires.

3.6.11 Hydraulic System. When a hydraulic hoist system is furnished, it shall show no leaks or breakage when the truck is subjected to the test prescribed in 3.6.10, with the pressure relief valve inoperative. This requirement does not include the pump.

3.6.12 Stability. Lateral stability of the unloaded truck shall be such that no overturning is induced when the load shoes are at maximum height and the platform on either side is raised to a 4 percent slope. This requirement will be measured in accordance with Test Method No. 9.

3.6.13 Acceleration. Trucks, when carrying rated load, shall be capable of accelerating from a standing start at an average rate of not less than 2.0 feet (610 mm) per second in the first 100 feet (3048 cm). Trucks, when unloaded, shall be capable of accelerating from a standing start at an average rate of not less than 4.3 feet (1310 mm) per second in the first 100 feet (3048 cm). Trucks may be started in second gear for purposes of this test. Compliance with this requirement shall be measured in accordance with Test Method No. 10.

3.6.14 Load Shoe Travel. The load shoe travel measured from the ground level to the load shoe carrying surface (with rated load) shall not be less than 12 inches (305 mm).

3.6.15 Clutch. The force required to depress the clutch pedal and disengage the clutch shall not exceed 50 pounds (23 kg) when measured in accordance with Test Method 11.

3.6.16. Truck Width. Overall truck width shall not exceed 101 inches (2565 mm) for 48 inch (1220 mm) wide pallet loads and 108 inches (2745 mm) for 54 inch (1370 mm) wide pallet loads.

3.6.17 Electromagnetic Interference Characteristics. Unless otherwise specified (see 6.2), the electromagnetic interference characteristics shall conform to SAE J551.

3.6.18 Noise Limits. The noise level of the straddle carry truck (excluding horns) shall not exceed 85 db(A) when tested under the following conditions:

(a) At the operator's station when tested in accordance with 4.3.2.6.
(b) At 24.5 feet (7470 mm) when tested in accordance with 4.3.2.6.

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The precautions of MIL-STD-1474 shall be provided if the noise level exceeds 85db(A) when tested in accordance with 4.3.2.6.

3.6.18.1 When closed cab is specified (see 6.2), the sound level at the operator's station shall not exceed 85db(A) with doors and windows in the closed position when tested in accordance with 4.3.2.6.

3.6.19 Torque Converter. The torque converter and transmission shall be capable of absorbing maximum stall operation for a minimum of 60 seconds without damage, permanent deformation, or exceeding the fluid temperature limit of 250°F (120°C) with an initial fluid temperature of not less than 157°F (69°C) in an ambient of not lower than 30°F (-1°C). This requirement will be measured in accordance with Test Method No. 13.

3.6.20 Performance. The truck shall successfully complete 1440 laps of the test course without failure. This requirement shall be measured in accordance with Test Method 14.

3.6.21 Drift. The hoist mechanism shall be capable of holding the rated load at maximum height not less than 15 minutes with not more than 2 inches of vertical drift. Drift rate shall be measured when hydraulic oil temperature is not less than 120 °F.

3.6.22 Visibility. The operator with the use of mirrors shall be capable of seeing the area directly in front of the truck to the point where operator's unobstructed visibility within the enclosed cab begins. Also, the operator shall be capable of seeing along both the truck's right and left side and to the rear of the truck. Finally, the operator shall have visibility capability within the enclosed cab to verify load position and engagement.

3.7 OTHER REQUIREMENTS

3.7.1 Bearings. Rotating parts shall be mounted on suitable antifriction bearings, wherever design permits, in accordance with the best automotive practice. Bearing load capacity, including applicable safety factors, as rated by the bearing manufacturer, shall be equal to or greater than maximum bearing loading under normal truck operation and under operation required by the test course.

3.7.2 Lubrication. Accessibility and means shall be provided for applying and where necessary checking the level of lubricant in all parts and components. A standard pressure grease lubrication system shall be provided for all friction points not lubricated from a reservoir. The truck shall be tested and furnished with standard military lubricants specified in Table V, FIG. 4, in the Appendix.

3.7.3 Workmanship. All workmanship on the truck shall be in accordance with the engineering and production standards of the material handling industry. The truck shall withstand any operation during normal work without permanent deformation, breakage of connections, malfunction, or component interference caused by incorrect workmanship. All parts of the truck before and after painting shall be clean and free from sand, rust, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

3.7.3.1 Steel and Other Metal Fabrication. Metals used in the fabrication of the truck shall provide original quality surface finish and shall be free

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from kinks and sharp bends. Metals having eroded or pitted surfaces are not acceptable. The straightening of materials shall be done by methods that will not cause weakening or injury to the material. Burrs and sharp edges in holes and on sheets, plates, and members shall be removed sufficiently to assure correct fits and to prevent loosening of fasteners and damage to components. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Splatter shall be removed from exposed cuts and from re-entrant cuts. Heated metals shall be allowed to cool slowly except during heat treatment, and over-heating shall be avoided in accordance with the recommendations of the metal manufacturer. All modular assembly fabrication shall provide for interchangeability of components.

3.7.3.2 Rivet Connections. Rivet holes, in the sizes recommended by standard practice, shall be accurately punched or drilled and shall have burrs removed. Rivets shall be driven with power tools. Rivet heads shall be full, neatly made, concentric with rivet holes, in full contact with the surface of the member, and in accordance with SAE J492. Excessive upsetting of rivets to fill holes will not be acceptable.

3.7.3.3 Bolted Connection. Bolt holes shall be accurately formed. Washers or lockwashers or a combination of the two shall be provided on all bolts, studs, and capscrews having straight thread. Self-locking nuts are acceptable in lieu of standard nuts and lockwashers. All nuts shall have full thread engagement, and all other threaded fasteners shall have tightening and thread engagement in accordance with SAE J475.

3.7.3.4 Welds. Welded joints shall be sound, smooth, and free from pits, holes, fissures, rough projecting edges, and slag. Each weld shall be thoroughly fused with the base metal.

3.7.3.5 Machine Work. All parts shall be manufactured for accurate assembly through the use of correct jigs, fixtures, or tape controlled machines or any desired combination of these. Like parts shall be interchangeable.

3.7.3.6 Reassembly. The contractor shall provide all services and personnel necessary for tear down, packaging, and reassembly at the destination of equipment items whose physical size prevent shipment in a fully assembled condition. Contractor shall inform the consignee in writing the truck has been shipped and upon being notified the truck has arrived at the activity the contractor will perform the reassembly operations within two weeks of written notification.

3.7.4 Treating and Painting. All exterior surfaces of the truck shall be thoroughly cleaned and shall be dry and free from mill scale, oil, grease, dirt and rust and shall be painted as soon as practicable after cleaning with not less than one coat of metallic base primer and two coats of synthetic enamel. Primer and finish coats shall be applied in accordance with normal commercial practices to assure complete coverage and durability of the finish. The finish color shall be contractor's commerical yellow unless otherwise specified (see 6.2) The finish coat when dry shall be a smooth even surface, free from runs, sags, peels, chips, blisters, and areas of no film. Surfaces of components and assemblies of the truck, not normally painted with a finish coat, shall be cleaned and painted in accordance with good commercial practices.

3.7.5 Identification Marking. The truck shall be identified with an identification plate conforming to MIL-P-514, type 1, style I, composition C which shall be in visible and safe location on the instrument panel or fire wall

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of the truck. The identifying data to be applied to the identification plates shall be as indicted herein and as defined in MIL-STD-130. The truck shall also be equipped with instruction, warning, and caution plates conforming to MIL-P-514, prominently located and describing any special or important procedure to be followed in operating, loading, and servicing of the truck or its components. All plates shall be securely attached to the truck with screws, bolts, or rivets, and shall be furnished and mounted by the suppliers.

TRUCK, STRADDLE-CARRY, DIESEL, PNEUMATIC RUBBER TIRES

CAPACITY.....pounds
NAME OF MANUFACTURER.....
SERIAL NUMBER.....
MODEL NUMBER.....
REGISTRATION NUMBER.....
DELIVERY DATE.....(month and year)
CONTRACT OR ORDER NUMBER.....
INSPECTED.....(name of inspection office)
SERVICE WEIGHT.....pounds .....(kg)
WHEEL LOADING (no load)
FRONT WHEELS (each).....pounds .....(kg)
REAR WHEELS (each).....pounds .....(kg)
(1) WHEEL LOADING (rated load)
FRONT WHEELS.....pounds .....(kg)
REAR WHEELS (each).....pounds .....(kg)
CENTER OF GRAVITY (no load)
HORIZONTAL....inches .....(mm) from axle of front wheels
VERTICAL.....inches.....(mm) from axle of front wheels
IDENTIFICATION OF MAINTENANCE MANUAL.....(Assigned by U.S. Government)

NOTE (1) Load in carrying position.

3.7.5.1 Shipping Data Plate. Shipping data plate shall conform to MIL-P-514, type III, composition C, shall be furnished and in addition shall show the si;houette of the truck in transport position indicating the center of gravity and the location and capacity of the lifting and tiedown attachments. The plate shall be attached by screws, bolts, orrivets in a conspicuous location. As a minmum the plate shall have the following information:

Wheel Loading (no load on shoes)
Drive Wheels (each) _____pounds
Steer Wheels (each) _____pounds

Wheel Loading (rated load on shoes)
Drive Wheels (each) _____pounds
Steer Wheels (each) _____pounds

3.7.6 Vehicle marking. The lifting capacity of the truck shall be painted in block numbers three inches (75mm) high on each side of truck. The assigned registration number for each truck shall be painted in block numbers three inches (75mm) high on each side of engine hood. Tire pressures for each tire shall be stenciled in numbers one inch high (25mm) on each side of the truck near the applicable tire. When applicable (see 6.2), durable black enamel markers indicating the safety designation of the truck shall be applied on each side and on the rear of the truck in a visible but protected location. These

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markers shall be in accordance with ANSI/UL558. In addition, supplementary marking and load handling symbols as specified in ANSI MH11.3 are also required. Slinging and tiedown markings shall be in accordance with MIL-STD-209. "No rider warning" shall be painted in block numbers two inches (50mm) high on side of truck. Fuel type shall be painted in block numbers one inch (25mm) high near fuel tank filler. Block numbers shall be painted in black enamel.

3.7.7 Repair and Maintenance. Provision shall be made for ready adjustment, service, or replacement of fan belt, ignition parts or assemblies, carburetor, fuel pump, oil filter, alternator, alternator regulator, battery, wearing parts of steering assembly, tires, wheels, lights, and horn. If access openings are used, each opening shall be provided with a removable or hinged cover closing against a rubber gasket around the periphery of the opening. There shall be no interference to the servicing or draining of oils and lubricants to or from any assembly or component by frame members or other obstructions.

3.7.8 Technical Manuals. Unless otherwise specified (see (6.2), one copy of the technical manual conforming to Specification MIL-M-21861 shall be furnished by the manufacturer with each truck at time of shipment. Additional copies of the manual shall be furnished as specified. The manuals are to be preserved and packaged in transparent plastic by Method IC-1 of MIL-P-116 with each bag heat sealed.

3.8 Interchangeability. Corresponding units and replaceable assemblies, subassemblies, and parts in contact shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the items are used. Where dimensions, ratings, tolerances, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. Whether subassemblies and parts are replaceable shall be determined by the contracting officer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction (First Produced) Testing. The preproduction sample truck, if required (see 6.2), shall be tested for all provisions of this specification. These tests shall be witnessed by the contracting officer or his authorized representative or appointed test officer. Failure of the preproduction sample to pass these tests shall be cause for rejection of the preproduction sample. If a change is made in design, material or construction, subsequent to the approval of the preproduction sample, retest shall be required. When specified (see 6.2), the supplier shall furnish certificate of compliances, as specified herein, certifying that the component complies in all particulars with the requirements of this specification.

4.1.1 Inspection Responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection and testing requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Acceptable criteria. Trucks that conform to all requirements in Section 3 and 5 of this specification and pass all applicable inspections tested

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in Section 4 of this specification will be considered acceptable by the Government.

4.1.3 Component and Material Inspection. The supplier is responsible for assuring that components and materials used are manufactured, examined, and tested in accordance with requirements of referenced specifications and standards.

4.2 Classification of inspections and test. Inspections and tests shall be classified as follows:

(a) First produced truck inspection and tests (see 3.1.1, 4.3 and 6.2).

(b) Post Test inspection and tests (see 4.4 and 6.2).

(c) Quality conformance inspection and tests for each production truck (see 4.5).

(d) Inspection comparison (see 4.6).

(e) Inspection of preparation for delivery (see 4.7).

4.3 First produced truck inspection and tests. Prior to inspection and test of the truck(s) specified in 3.1.1 and 6.2 the following shall be performed:

(a) Furnish a schedule of maintenance to be followed during all testing of the first produced truck. A list of tools required to perform this maintenance shall also be provided.

(b) Service the truck with oils and greases specified herein and designated for use in the ambient temperature at which the tests will be conducted.

(c) Break in the truck as prescribed by the supplier.

4.3.1 Inspections

4.3.1.1 Pretest inspection. Prior to testing according to 4.3.2, one or more of the first produced trucks as specified in 3.1.1 and 6.2 shall be inspected for conformance marked "X" in Column 1 of Table III.

4.3.2 Tests. Upon sucessful completion of the inspection specified in 4.3.1.1, the truck shall be subjected to the tests marked "X" in Column 1 of Table IV and as specified herein. Acceptance of a first produced truck shall not exclude the remaining trucks from the quality comformance inspection and acceptance provisions specified in Section 4. Test conditions and schedule shall be as follows:

(a) Unless otherwise specified herein, tests shall be conducted at the ambient temperature and climatic conditions existing at the place of test. The test surface, unless otherwise specified or specific tests shall be level within + 1%. Only that maintenance established by the supplier and submitted as a maintenance schedule prior to commencement of tests shall be performed during the tests.

(b) Tests shall be in accordance with Table IV and may be conducted in any order desired except that reliability and post test shall be the last tests

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conducted. An "X" in the applicable column indicates the test that shall be conducted.

(c) Inspection and tests requiring use of test loads shall utilize, as applicable, a palleted or skidded, adequately strapped, uniformly distributed load of the following characteristics:

Maximum Load Length - See 6.2 for required shoe length for Sizes 1 thru 8. Use longest shoe length for test.

Maximum Height of Load Including Pallet - 108 inches (2475 mm) for Code A and E
- 96 inches (2440 mm) for Code B and F
- 80 inches (2030 mm) for Code C and G
- 60 inches (1525 mm) for Code D and H

Maximum Width of Load - 48 inches (1220 mm) for Sizes A. B. C. and D
- 54 inches (1370 mm) for Sizes E, F, G, and H

Rated load - 30,000 pounds (13,605 kg) for all sizes

The center of gravity of the load shall be on the centerline of the wheelbase and the centerline of the truck. The vertical center of gravity shall be no higher than the vertical center of the load space. A tolerance of plus or minus one percent of the specified test load will be permitted.

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TABLE III. Inspection Schedule

1 First Produced	2 Quality Conformance	3 Post Inspec	4 Requirements as Specified	5 Yes	6 No
X	-------	----	Safety (see 3.1.3.1)		
X	-------	----	Filling, Draining and Checking Provisions (see 3.1.3.2.1)		
X	-------	----	Alternator (see 3.2.2.1)		
X	-------	----	Alternator Regulator (see 3.2.2.2)		
X	-------	----	Starting System (see 3.2.2.3)		
X	-------	----	Fuel System (see 3.2.2.4)		
X	-------	----	Air Cleaner (see 3.2.2.4.1)		
X	X	----	Cooling System (see 3.2.2.5)		
X	-------	----	Fan (see 3.2.2.5.1)		
X	-------	----	Water Pump (see 3.2.2.5.2)		
X	-------	----	Thermostat (see 3.2.2.5.3)		
X	-------	----	Radiator (see 3.2.2.5.4)		
X	-------	----	Gaskets (see 3.2.2.5.5)		
X	-------	----	Governor (see 3.2.2.6)		
X	-------	----	Exhaust System (see 3.2.2.7)		
X	-------	----	Spark Arresting Device (see 3.2.2.7.1)		
X	-------	----	Cold Weather Starting System (see 3.2.2.8)		
X	-------	----	Drive Assembly (see 3.2.3)		
X	-------	----	Transmission (see 3.2.3.1, 3.2.3.1.1 and 3.2.3.1.2)		
X	-------	x	Torque Converter (see 3.2.3.2)		
X	-------	----	Transmisssion Controls (see 3.2.3.3)		
X	-------	x	Drive Shaft (see 3.2.3.4.1)		
X	-------	----	Differential (see 3.2.3.4.2)		
X	-------	----	Drive Axles (see 3.2.3.4.3)		
X	-------	x	Drive Head and Chains (see 3.2.3.4.4)		
X	-------	----	Power take-off (see 3.2.4.1.1)		
X	-------	----	Take-off Shafts (see 3.2.4.1.2)		
X	-------	----	Hoist Brake (see 3.2.4.1.3)		
X	-------	----	Hoist and Lower Gear Boxes (see 3.2.4.1.4)		
X	-------	----	Side Shafts (see 3.2.4.1.5)		
X	-------	----	Shoe Lift Rods (see 3.2.4.1.6)		
X	-------	----	Limit Stop (see 3.2.4.1.7)		
X	-------	----	Load Shoes (see 3.2.4.1.8)		
X	-------	x	Hydraulic System (see 3.2.4.2)		
X	-------	----	Hydraulic Cylinders (see 3.2.4.2.1)		
X	-------	----	Hydraulic Lines (see 3.2.4.2.2)		
X	-------	----	Hydraulic Valves (see 3.2.4.2.3)		
X	-------	----	Reservoir (see 3.2.4.2.4)		
X	-------	----	Lever Arms (see 3.2.4.2.5)		
X	-------	----	Frame (see 3.3.1)		
X	-------	----	Wheels (see 3.3.2)		
X	-------	x	Tires (see 3.3.2.1)		
X	-------	----	Tubes (see 3.3.2.2)		
X	-------	x	Steering (see 3.3.3)		

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TABLE III. Inspection Schedule

1 First Produced	2 Quality Conformance	3 Post Inspec	4 Requirements as Specified	5 Yes	6 No
X	———	———	Wheel Mounting (see 3.3.3.1)		
X	———	———	Wheel Forks and Tubes (see 3.3.3.2)		
X	———	x	Springs (see 3.3.3.3.)		
X	———	x	Service Brakes (see 3.3.4.1)		
X	———	———	Parking Brake (see 3.3.4.2)		
X	———	———	Body (see 3.3.5)		
X	———	———	Seat (see 3.3.5.1)		
X	X	———	Closed Cab (see 3.3.5.2)		
X	———	———	Cowl or Hood (see 3.3.5.3)		
X	———	———	Windshield (see 3.3.5.4)		
X	———	———	Instrument Panel (3.3.5.5)		
X	———	———	Slinging and Tie Provisions (see 3.3.5.6)		
X	———	———	Hand Holes (see 3.3.5.7)		
X	———	———	Controls and Instrumentation (see 3.4)		
X	———	———	Controls (see 3.4.1)		
X	———	———	Light Switch (see 3.4.1.1)		
X	———	———	Engine Run Control (see 3.4.1.2)		
X	———	———	Starting Motor Switch (see 3.4.1.3)		
X	———	———	Brake Pedal (see 3.4.1.4)		
X	———	———	Accelerator Control (see 3.4.1.5)		
X	———	———	Hoist and Shoe Swing Contols (see 3.4.1.6)		
X	———	———	Loud Light Switch (see 3.4.1.7)		
X	———	———	Clutch Pedal (see 3.4.1.8)		
X	———	———	Instruments (see 3.4.2)		
X	———	———	Ammeter or voltmeter (see 3.4.2.1)		
X	———	———	Oil Pressure Guage (see 3.4.2.2)		
X	———	———	Temperature Guage (see 3.4.2.3)		
X	———	———	Hour Meter (see 3.4.2.4)		
X	———	———	Speedometer (see 3.4.2.5)		
X	———	———	Fuel Guage (see 3.4.2.6)		
X	———	———	Direction Indicators (see 3.4.2.7)		
X	———	———	Headlights (see 3.5.1)		
X	———	———	Stoplights (see 3.5.2)		
X	———	———	Wiring and Switching (see 3.5.2.1)		
X	———	———	Interlocking Light Switch (see 3.5.2.2)		
X	———	———	Load Lights (see 3.5.3)		
X	X	x	Battery (see 3.5.4)		
X	———	———	Battery Mounting (see 3.5.4.1)		
X	———	———	Horn (see 3.5.5)		
X	———	———	Safety Guards and Platforms (see 3.5.6)		
X	X	———	Fire Extinguisher (see 3.5.7)		
X	———	———	Dual Rearview Mirrors (see 3.5.8)		
X	———	———	Guardrails (see 3.5.9)		
X	———	———	Safety Features (see 3.5.10)		
X	———	———	Bearings (see 3.7.1)		
X	X	———	Lubrication (see 3.7.2)		

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TABLE III. Inspection Schedule

1	2	3	4	5	6
First Produced	Quality Conformance	Post Inspec	Requirements as Specified	Yes	No
X	X	----	Workmanship (see 3.7.3)		
X	X	----	Steel and othe Metal Fabrications (see 3.7.3.1)		
X	X	----	Rivet Connections (see 3.7.3.2)		
X	X	----	Bolted Connection (see 3.7.3.3)		
X	X	----	Welds (see 3.7.3.4)		
X	-------	----	Machine Work (see 3.7.3.5)		
X	X	----	Treating and Painting (see 3.7.4)		
X	X	----	Identification Marking (see 3.7.5)		
X	X	----	Shipping Data Plate (see (3.7.5.1)		
X	X	----	Vehicle Marking (see 3.7.6)		
X	X	----	Technical Manuals (see 3.7.8)		

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TABLE IV. Test Schedule

1	2	3	4	5	6	7
First Produced	Quality Conformance	Post Tests	Test	Test Paragraph	Performance Paragraph	Test Method
X	———	——	Lifting Speed	4.3.2.1	3.6.1	1
X	———	——	Lowering Speed	4.3.2.1	3.6.2	1
X	———	——	Turning Radius	4.3.2.2	3.6.3	———
X	———	X	Steering	4.3.2.3	3.3.3 and 3.6.4	2
X	X	——	Speed	4.3.2.4	3.6.5	3
X	X	X	Slope Ascension	4.3.2.5	3.6.6	4
X	———	——	Underclearance	4.3.2.6	3.6.7	14
X	X	X	Service Brakes	4.3.2.7	3.6.8	5,6,7
X	———	——	Parking Brake	4.3.2.8	3.3.4.2	14
X	X	——	Lift Assembly	4.3.2.9	3.6.9	———
X	X	——	Frame, Wheel, Forks, Tubes & Hoist Mech.	4.3.2.10	3.6.10	8
X	———	X	Hydraulic System	4.3.2.11	3.6.11	8
X	———	——	Stability	4.3.2.12	3.6.12	9
X	———	——	Acceleration	4.3.2.13	3.6.13	10
X	———	——	Load Shoe Travel	4.3.2.14	3.6.14	———
X	———	X	Clutch	4.3.2.15	3.6.15	11
X	———	——	Truck Width	4.3.2.16	3.6.16	———
X	———	——	Electromagnetic Interference Chacteristics	4.3.2.17	3.6.17	———
X	———	——	Noise	4.3.2.18	3.6.18	———
X	———	——	Low Temperature	4.3.2.19	3.1.5 and 3.2.2.8	12
X	———	X	Torque Converter	4.3.2.20	3.6.19	13
X	———	——	Slinging and tiedown Attachments	4.3.2.21	3.3.5.6	——— ———
X	———		Performance	4.3.2.22	3.6.20	14
X	———	X	Drift	4.3.2.23	3.6.21	15
X	X	——	Visibility	4.3.2.24	3.6.22	-16—

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4.3.2.1 Lifting speed and lowering speed. Test the truck in accordance with Test Method No. 1. Nonconformance to 3.6.1 and 3.6.2 shall constitute failure of this test.

4.3.2.2 Turning Radius. Test the truck to determine compliance with the requirements of 3.6.3. Contractor shall lay out a test course on a paved area to simulate a 90° right turn intersecting aisle. Movable vertical barriers shall be placed at the inside and outside corners to define the aisle width. Any movement of the vertical barriers caused by the truck shall constitute failure of this test. Trucks shall turn in one motion, without backing, and to meet this requirement in no more than three trials. Repeat test for 90° left turn intersecting aisle.

4.3.2.3 Steering. Test the truck in accordance with Test Method No. 2. Nonconformance to 3.3.3 and 3.6.4 shall constitute failure of this test.

4.3.2.4 Speed. Test the truck in accordance with Test Method No. 3. Nonconformance to 3.6.5 shall constitute failure of this test.

4.3.2.5 Slope ascension. Test the truck in accordance with Test Method No. 4. Nonconformance to 3.6.6 shall constitute failure of this test.

4.3.2.6 Underclearance. Test the truck in accordance with Test Method No. 14. Nonconformance to 3.6.7 shall constitute failure of this test.

4.3.2.7 Service brakes. Test the truck in accordance with Test Method No. 5,6 and 7. Nonconformance to 3.6.8 shall constitute failure of this test.

4.3.2.8 Parking Brake. Test the truck in accordance with Test Method No. 14. Nonconformance to 3.3.4.2 shall constitute failure of this test.

4.3.2.9 Lift Assembly. Test the truck to determine compliance with 3.6.9. The rated load shall be lifted to 12 inches (305 mm) measured from the ground level to the load shoe carrying surface. Appreciable wear is defined as any apparent inability to lift and hold rated load for a period of not less than 30 minutes with not more than 1/4 inch (6 mm) vertical drift for hydraulic system and no drift for mechanical system.

4.3.2.10 Frame, wheel, forks, tubes and hoist mechanism. Test the truck in accordance with Test Method No. 8. Nonconformance to 3.6.10 shall constitute failure of this test.

4.3.2.11 Hydraulic system. Test the truck in accordance with Test Method No. 8. Nonconformance to 3.6.11 shall constitute failure of this test.

4.3.2.12 Stability. Test the truck in accordance with Test Method No. 9. Nonconformance to 3.6.12 shall constitute failure of this test.

4.3.2.13 Acceleration. Test the truck in accordance with Test Method No. 10. Nonconformance to 3.6.13 shall constitute failure of this test.

4.3.2.14 Load Shoe Travel. Test the truck to determine compliance with 3.6.14. Truck shall be placed on level paved surface and each load shoe shall be leveled with a 8 foot mason's level. Measure distance from paving to load shoe carrying surface.

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4.3.2.15 Clutch. Test the truck in accordance with Test Method No. 11. Nonconformance to 3.6.15 shall constitute failure of this test.

4.3.2.16 Truck Width. Measure the overall truck width from the outside to outside wheel guard dimension to ensure compliance with 3.6.16.

4.3.2.17 Electromagnetic Interference Characteristics. The truck shall be tested to ensure compliance with 3.6.17. The supplier shall furnish the contracting officer the report of the test . Upon approval of the report by the contracting officer, the approved truck shall be used as a model for all production trucks. No substitute of any truck component effecting the EMI test results will be allowed.

4.3.2.18 Noise Level measurement

4.3.2.18.1 Operator's Station. Measure equivalent continuous noise level (Leq) in accordance with MIL-STD-1474A at the operators station during performance testing (Test Method No. 14). Duty cycle test shall be run in lieu of the steady-state test of MIL-STD-1474A. Minimum speed of negotiating the course shall be as specified in 3.6.5. All lifting and lowering operations shall be accomplished with the truck stationary at lifting and lowering speeds specified. Cumulative noise exposure measurement shall require a minimum of 1 hour of continuous operation on the test course.

4.3.2.18.2 Noise Hazard. When the noise level of the truck exceeds 85db(A) (see 3.6.18) when tested in accordance with 4.3.2.18.1 noise hazard signs conforming to MIL-STD-1474 shall be posted by the contractor on each side and rear of truck.

4.3.2.18.3 Noise Hazard. When a closed cab is specified (see 6.2), the maximum noise level at the operator's station (microphone located no more tha six 6 inches from operator's approximate ear location) shall not exceed 85db(A) (see 3.6.18.1). Noise level in excess of 85db(A) shall constitute failure of this test.

4.3.2.19 Low temperature. Unless otherwise specified (see 6.2), test the truck in accordance with Test Method No. 12. Nonconformance to 3.1.5 and 3.2.2.8 shall constitute failure of this test.

4.3.2.20 Torque convertor. Test the truck in accordance with Test Method No. 13. Nonconformance to 3.6.19 shall constitute failure of this test.

4.3.2.21 Slinging and tiedown attachments. Test the slinging provisions (see 3.3.5.6) by applying the load for the time period specified in MIL-STD-209. Test the tiedown provisons by applying the ultimate load for the time period specified in MIL-STD-209 (see 3.3.5.6).

4.3.2.22 Performancee. Test the truck in accordance with Test Method No. 14. Nonconformance to 3.6.20 shall constitute failure of this test.

4.3.2.23 Drift. Test the truck in accordance with Test Method No. 15. Nonconformance to 3.6.21 shall constitute failure of this test.

4.3.2.24 Visability. Test the truck in accordance with Test Method No. 16. Nonconformance to 3.6.22 shall constitute failure of this test and the contractor will be required to supply additional mirror(s) in order to demonstrate compliance.

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4.4 Post Test Inspections and Tests.

4.4.1 Post Test Inspections. After sucessful completion of testing specified in 4.4.2, the first produced truck(s) shall be inspected for compliance marked "x" in column 3 of Table I.

4.4.2 Post Tests. The first produced truck(s) shall be subjected to the test marked "x" in column 3 of Table II.

4.5 Quality Comformance Inspection and Tests.

4.5.1 Quality Conformance Inspections. Upon successful completion of all tests specified in 4.5.2, each production truck shall be inspected for compliance marked "x" in column 2 of Table I.

4.5.2 Quality Conformance Tests. Test each production truck for compliance marked "x" in column 2 of table II.

4.6 Inspection Comparison. The government may select trucks at any time during the contract production period and subject the trucks to the inspections specified in Column 1 of Table I and to the tests specified in Column 1 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected and tested truck(s). In addition to any test specified as part of the inspection comparison, the government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure ot those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison truck to meet any requirement specified herein during and as a result of the inspection and tests specified herein during and as a result of the inspection and tests specified in 4.2 shall be cause for rejection of the inspection comparison truck(s) and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all trucks accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.7 Preproduction sample truck(s). Any changes or deviations of production truck(s) from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish truck(s) conforming to this specification (see 3.1.1).

4.8 Incident report. When the contractor conducts the demonstration and test specified herein, a written report within 24 hour, of any malfunction or failure during the conduct of the demonstration and test is required. As a minimum, the report shall describe components and parts affected, test and operating conditions, date of incident, hour meter reading, how detected and description

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of incident and corrective action taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packing, and marking. Each complete straddle carry truck shall be preserved, packed and marked in accordance with MIL-STD-162 for type I mobile or type II crated, as specified (see 6.2). The degree of preservation shall be level A, Level B, Level C or commercial, and the degree of packing shall be level A, level B, level C or commercial as specified (see 6.2).

6. NOTES

6.1 Intended Use. The trucks described herein are intended for moving cargo in yards, around warehouses, and terminals, and over-the-road within the military supply system. The trucks are intended for operation over paved surfaces for long distances.

6.2 Ordering data, procurement documents should specify the following:

(a) Title, number and date of the specification.

(b) List of size code(s) required (see 1.2).

(c) When special length of load shoes, other than those specified, are required (see 1.2 and 3.2.4.1.8).

(d) When preproduction sample truck is required, number and size of trucks to be furnished, and time frame required for submission of preproduction test results (see 3.1.1).

(e) List of required certificates of compliance (see 3.1.4) and testing laboratory certification (see 3.1.3.1)

(f) When alternate truck operating temperatures are required (see 3.1.5).

(g) When coolant in coolant system is not required (see 3.2.2.5).

(h) When "DS" safety rating and backfire deflector are required (see 3.1.3.1 and 3.2.2.7.1).

(i) When powershift transmission is required (see 3.2.3.1 and 3.2.3.1.1.).

(j) When drive assembly torque test is required (see 3.2.3).

(k) When closed cab is not required (see 3.3.5.2).

(l) When cab heater is not required with closed cab (see 3.3.5.2).

(m) When battery condition shall be wet and charged in lieu of dry and charged (see 3.5.4).

(n) When fire extinguisher is required (see 3.5.7).

(o) When electromagnetic interference charcteristics shall conform to MIL-STD-461 in lieu of SAE J551(see 3.6.17).

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(p) When technical manual need not conform to MIL-T-21861 (see 3.7.8).

(q) When low temperature test is not required (see 3.1.5).

(r) When other than contractor's commerical yellow finish is required, specify type (see 3.7.4).

(s) Type and degree of preservation, packaging and packing required (see 5.1).

Custodians:

Army - ME
Navy - SA
Air Force - 99

Review activities:

Army - SM, MT
Navy - MC,SH
DLA - CS
Air Force -84

User activity:

Navy - SH

Preparing activity:

Navy - SA

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APPENDIX

FOR

TRUCKS, STRADLE-CARRY, 30,000 POUNDS (13,605 kg) CAPACITY, PNEUMATIC RUBBER TIRES

1.0 Scope. The Test Methods contained within this Appendix are to determine Whether trucks procured under this specification conform to the requirements Listed herein.

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TEST METHOD NO. 1

LIFTING AND LOWERING SPEEDS

1. Purpose: To determine that the lifting and lowering speeds comply with the specified requirements.

2. LIFTING SPEED

2.1 Test Description:

2.1.1 This test shall be conducted with the truck straddling the rated load. The load shoes shall be in the fully lowered position.

2.1.2 Measure and mark a point of reference which indicates the load has been raised 12 inches (305 mm).

2.1.3 Commence the lifting operation and by use of a split second timing device determine the time required to contact the load and rise the load 12 inches (305 mm).

2.1.4 The load shall then be lowered and the above operation repeated three (3) times.

3. LOWERING SPEED

3.1 Test Description:

3.1.1 This test shall be performed with completely unloaded shoes at the 12 inch (305 mm) reference point marked in 2.1.2.

3.1.2 Commence lowering the shoes and by use of a split second timing device shall determine the time required to lower the shoes and swing them out to the maximum spread position.

3.1.3 The unloaded shoes shall be returned to the initial positon and the above operation repeated three (3) times.

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TEST METHOD NO. 2

STEERING

1. Purpose. To determine whether the force required to turn the hand steering wheel meets specifications, the number of turns necessary to turn the wheels from extreme right to extreme left, to determine the amount of free play and whether the steering system can withstand without permanent damage, the maximum specified force with steerable wheels blocked.

2. STEERING FORCE TEST

2.1 Test Description. Place the unloaded truck on a level paved surface. Using a tension dynamometer (with a 0-300 pound (0-136 kg) dial) or equivalent, equipped with a maximum hand, measure the force required to move the steering wheel.

2.2 Test Procedure:

2.2.1 Place truck on dry, level paved surface.

2.2.2 Place steer wheels in a straight-ahead position.

2.2.3 Assemble an adapter pulley having a root diameter equal to the steering wheel diameter and means for attachment to the wheel spokes of the hand steering wheel.

2.2.4 Attach a tension dynamometer to the adpter pulley.

2.2.5 Grasp the dynamometer lead and apply a steady force, in a counterclockwise direction and in the wheel plane, until the wheel turns.

2.2.6 Observe the maximum gage reading within the first half of the inner steer wheel angle.

2.2.7 Observe the maximum gage reading at extreme ends of the steer.

3 . UNBALANCED TANGENTIAL FORCE

3.1 Block steering wheels in a straight-ahead position.

3.2 Grasp steering wheel and exert at least 150 pounds (68 kg) pressure in clockwise direction.

3.3 Repeat in counterclockwise direction.

3.4 Perform operations 2 and 3 four times.

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TEST METHOD NO. 2 CONTINUED

4. STEERING WHEEL TURNS

4.1 Count the number of turns of the hand steering wheel required to turn the four wheels from extreme right position to the extreme left.

4.2 Repeat operation from extreme left to extreme right.

5. STEERING WHEEL FREE PLAY

5.1 Test Description: With the truck stationary and with no load on the lift shoes, the amount of free play in the steering wheel is measured. This procedure is followed before and after the performance test. If 1/4 inch (25 mm) increase in free play is noted after the performance test (Test Method No. 14) is run, it is an indication of excessive wear in the steering mechanism and constitutes failure of this test.

5.2 Test Procedure:

5.2.1 Place unloaded truck on a dry, level, paved surface with its rear wheels parallel to the longitudinal axis of the truck.

5.2.2 Attach to the steering column, immediately below the steering wheel and parallel to it, a piece of stiff fiber board of similar material which is semi-circular in shape and which has a radius at least 1 inch greater than the radius of the steering wheel

5.2.3 Mark a reference point on the upper portion of the outer edge of the steering wheel; mark a corresponding point on the fiber board templet.

5.2.4 Turn the steering wheel counterclockwise until the rear wheels of the truck begin to turn. Mark a point on the fiber board templet corresponding to the reference point on the outer edge of the steering wheel.

5.2.5 Turn the steering wheel clockwise until the rear wheels of the truck begin to turn. Mark a point on the fiber board templet corresponding to the reference point on the outer edge of the steering wheel.

5.2.6 Remove the fiber board templet from the truck.

5.2.7 Place the fiber board templet on a flat surface. Measure the distance between the 2 points made in accordance with operations 5.2.4 and 5.2.5 above. This distance, in inches or millimeters, is the free play of the steering wheel.

5.2.8 Repeat operations 5.2.1, 5.2.2, 5.2.3, 5.2.4, 5.2.5, 5.2.6 and 5.2.7 after the performance test is run.

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TEST METHOD NO. 3

SPEED

1. Purpose: To determine whether truck, with rated load, on a level surface is capable of a forward speed in high gear of at least 25 miles per hour (11.2 m/sec.) and reverse speed in high gear of at least 17 miles per hour.

2. Test Description: Test is conducted on a 2676 foot (815 m) paved level course. Truck, with rated load, is accelerated through a 2200 foot (670 m) approach, timed through a 176 foot (54 m) test length, and then brought to a stop. Three runs are made forward and reverse in high gear. The data is recorded and the average speed determined.

3. Course Layout:

A — 2,200' (670 m) — B — 176' (54 m) — C — 300' (91 m) — D

4. Test Procedure:

4.1 Warm up truck for 15 minutes prior to test.

4.2 Locate truck, carrying rated load, for forward run with front wheels on line A.

4.3 Start test run and have truck in high gear and at speed by the time it reaches line B.

4.4 Using split second stop watch or electronic timing device, determine time in seconds used in traveling from line B to line C and record data.

4.5 Stop truck between line C and line D.

4.6 Perform operations 4.2, 4.3, 4.4, and 4.5 a second and third time.

4.7 Locate truck carrying rated load, for rearward run with rear wheels on line A.

4.8 Perform operations 4.3, 4.4, and 4.5 a second and third time.

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TEST METHOD NO. 4

SLOPE ASCENSION

1. Purpose: To determine whether truck when carrying rated load can accelerate from a dead stop on a 10 percent grade.

2. Test Description: Truck, carrying rated load, ascends a 10 percent grade in a forward direction on dry paved surface. Truck stops and starts up slope again.

3. Test Procedure:

3.1 Place rated load on load shoes.

3.2 Drive truck up 10 percent ramp in a forward direction and bring it to a dead stop.

3.3 Start Truck from the dead stop and proceed up ramp.

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TEST METHOD NO. 5

SERVICE BRAKE PRESSURE SENSITIVITY

1. Purpose: To determine whether a brake pedal pressure within the range of 50 to 150 pounds (68 kg) will develope a drawbar pull equal to 30 per cent of the gross weight of the truck (with rated load) while truck is being pulled.

2. Test Description: Pull fully loaded truck over dry, paved surface, with engine disengaged and parking brake released. Using a Bennett Feragen Corp. "Press-O-Meter", Model PM-35 (with a 0-200 pound (0-(90 kg) dial) or the equivalent, equipped with a maximum hand and suitable clamps, and a tension dynamometer fixed to the truck. If the paved surface does not have sufficient coefficient of friction, the surface may be brushed with Minnesota Mining Compound or other material to provide suitable coefficient of friction to prevent wheels from sliding while applying the required draw bar pull.

3. Test Procedure:

3.1 Record total weight of truck plus rated load.

3.2 Attach pressure gage to brake pedal, by means of clamps, in a manner which will allow the operator to apply foot pressure to one face of the gage.

3.3 Attach a tension dynamometer to the truck by means of a heavy steel cable.

3.4 Attach one end of a block and tackle, or similar device rendering a mechanical advantage, to the tension dynamometer, and the other end to an anchoring device.

3.5 With transmission in neutral, apply sufficient brake pedal pressure to prevent wheels from turning.

3.6 Apply tension to the dynamometer until force is equal to 30 percent of the total weight of truck plus rated load.

3.7 Gradually reduce brake pedal pressure until wheels just begin to turn while maintaining the dynamometer tension. Then pedal pressure shall not be less than 50 pounds nor more than 150 pounds (68 kg).

3.8 Record brake pedal pressure and dynamometer readings.

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TEST METHOD NO. 6

SERVICE BRAKE PRESSURE

1. Purpose: To determine whether brake pedal and brake system have the strength required by the specifications.

2. Test Description: Using a Bennett Feragen Corp. "Press-O-Meter", Model PM-35 (with a 0-300 pound (0-136 kg) dial) or the equivalent, equipped with a maximum hand and suitable clamps, apply a pressure of 250 pounds (115 kg) to the brake pedal 10 times.

3. Test Procedure:

3.1 Attach the pressure gage to the brake pedal by clamp to permit application of full foot pressure to one side of the pressure gage.

3.2 Apply foot pressure to the gage until it registers 250 pounds (115 kg) and hold for 1 minute.

3.3 Repeat operation number 3.2, 10 times.

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TEST METHOD NO. 7

STOPPING DISTANCE

1. Purpose: To determine if truck can be stopped within a given distance from specified speeds.

2. Test Description: Run truck with rated load in carry position 25 miles per hour (11.2 m/s) over dry, paved surface. Using a Bennett Feragen Corp. "Press-O-Meter", Model PM-35 (with a 0-200 pound (0-90 kg) dial) or the equivalent, apply specified brake pedal pressures and measure the distance required to stop the truck.

3. Course Layout:

A — 100' (30 m)or more — B — 88' (27 m) — C

4. Test Procedure:

4.1 Place a clearly visible vertical mark on the side of the truck directly beneath the normal position of the operator's head. Mark shall be such that the operator can see it easily.

4.2 Warm up truck for 15 minutes prior to test. Operate brakes at least 10 times.

4.3 Attach pressure gage to brake pedal by means of clamps which will allow the operator to apply foot pressure to one face of the gage.

4.4 Run the unloaded truck in forward direction onto a strip of dry, measured and marked as specified in "Course Layout".

4.5 Start two split-second stop watches or electronic timing devices at same time so that they are synchronized.

4.6 Station an observer at line B and a second observer at line C. Each oberserver shall be provided with one of the stop watches or electronic timing device described in operation 4.5.

4.7 Start the truck at line A and accelerate it so that it has attained a speed of 25 miles per hour (11.2 m/s) before it reaches line B.

4.8 Maintain the truck at a speed of 25 miles per hour (11.2 m/s) between lines B and C.

4.9 As the front wheels of the truck pass over line B, the first observer stops his watch.

4.10 As the front wheels of the truck pass over line C, the second observer stops his watch or timing device.

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TEST METHOD NO. 7 CONTINUED

4.11 The operator applies truck brakes, with a pedal pressure between 60 and 90 pounds (27 and 41 kg), when the line placed on the truck in operation 4.1 is directly over line C. The truck engine is not disengaged.

4.12 After truck is stopped, measure the distance from line C to the line on the truck and record data.

4.13 Record watch readings data.

4.14 Record maximum pedal pressure.

4.15 Perform operations 4.4 through 4.14 a second and third time.

4.16 Perform operations 4.4 through 4.15 with truck carying rated load. Load shall be in carrying position.

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TEST METHOD NO. 8

TRUCK FRAME, WHEEL FORKS, TUBES AND HOIST MECHANISM

1. Purpose: To determine whether the truck frame, hoisting mechanism, wheel forks, tubes and the related components can support a load equal to 200 percent of the rated capacity in the transport position with no signs of permanent deformation or failure.

2. Test Description:

2.1 This test shall be conducted with the truck straddling the bolster. The bolster plus the load it supports shall equal 200 percent of the rated load.

2.2 The load shoes shall be at the transport height in the transversely open position in such manner to allow the bolster containing the 200 percent of rated capacity to be raised by the use of lifting jacks or a hoisting system to an elevation slightly higher than the transport height.

2.3 The load shoes shall then be closed and retained against vertical movement in preparation to receive the overload.

2.4 The bolster shall then be slowly lowered on the jacks until bearing entirely on the load shoes.

2.5 This load shall be applied for 2 minutes then removed by the auxiliary lifting means.

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TEST METHOD NO. 9

LATERAL STABILITY

1. Purpose: To determine whether truck has sufficient lateral stability.

2. Test Description. The unloaded truck is tilted to one side on a platform which has a 4 percent slope. The operation is repeated with truck tilted to the opposite side. The truck has sufficient lateral stability if all its tires remain in contract with the platform under both conditions.

3. Test Procedure:

3.1 Drive unloaded truck onto platform which slopes 4 percent upward to the left.

3.2 Raise unloaded load shoes to maximum lift height.

3.3 Determine whether all tires remain in contact with platform.

3.4 Repeat operations 3.1, 3.2 and 3.4 while tilting truck to other side.

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TEST METHOD NO. 10

ACCELERATION

1. Purpose: To determine acceleration of truck, with and without rated load, on a level surface from a standing start.

2. Test Description: Test is conducted on a 300-foot (91 m) paved level course. Truck is accelerated at maximum rate from a standing start, timed through a distance of 100 feet (30 m), and then brought to a stop.

3. Course Layout:

A — 100' (30 m) — B — 200' (60 m) — C

4. Test Procedure:

4.1 Warm up truck for 15 minutes prior to test.

4.2 Locate truck, without load, with front wheels on line A.

4.3 Station an observer at line B. The observer shall be provided with a split-second stop watch or electronic timing device.

4.4 An observer leader shall be stationed at line A.

4.5 Observer shall start by calling "mark". When this signal is given, the truck operator starts the truck and the observer starts his watch. The truck may started in second gear for this test.

4.6 The truck operator will maintain maximum truck acceleration through the entire 100 foot (30 m) test run.

4.7 As the front wheels of the truck pass over line B, the observer stops his watch.

4.8 Record stop watch times.

4.9 Perform operations 4.1 through 4.8 a second and third time.

4.10 Perform operations 4.1 through 4.9 with truck carrying rated load. Load shall be in carrying position.

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TEST METHOD NO. 11

CLUTCH PEDAL

1. Purpose: To determine whether the force required to depress the clutch pedal, meets specifications.

2. Test Description: Using a Bennett Feragen Corp. "Press-O-Meter", Model PM-35 (with a 0-200 pound (0-90 kg) dial) or the equivalent, equipped with a maximum hand, measure the pressure required to depress the clutch pedal.

3. Test Prcedure:

3.1 Depress clutch pedal several times to insure proper lubrication.

3.2 Attach gage to clutch pedal to measure pressure required to depress pedal.

3.3 Apply pressure slowly and evenly until clutch pedal is depressed to 1 inch (25 mm) from the floor of the vehicle. Observe maximum gage reading and record data.

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TEST METHOD NO. 12

LOW TEMPERATURE

1. Purpose. To determine whether any damage has occurred to the truck due to storage at specified temperatures, engine will start within 5 minutes after beginning the first cranking, all instruments and gauges operate, battery charged system is operating showing a positive charge and truck and engine are operating smoothly and as specified.

2. Test Procedure:

2.1 Service the truck with applicable lubricants for specified (see 6.2) operation including engine oil, transmission oil, drive train lubricants, hydraulic oil and antifreeze.

2.2 Check the specific gravity of each battery to insure full charge.

2.3 Subject the truck to specified temperature until all parts, lube oil, coolant, and hydraulic oil have stabilized at specified temperature. Perform the test after stabilization is assured by temperature readings of plus or minus one half degree of specified temperature from thermo-couples located as follows:

2.3.1 In the two center cells of each battery between the plates and at middepth of the electrolyte.

2.3.2 In the volumetric center of the hydraulic system oil reservoir.

2.3.3 In the engine lubricating oil gallery and in the oil sump.

2.3.4 In the engine coolant in the engine-cooling-jacket discharge pipe.

2.3.5 In the transmission system oil reservoir.

2.4 Maintain specified ambient temperature with $\pm$ 2 ^{o}F ($2_{o}C$) throughout the entire test.

2.5 One operator wearing cold weather clothing, including mittens, shall perform all priming and starting operations.

2.6 Crank the engine using batteries that were cold soaked and stabilized at specified temperatures.

2.7 Substitute or supplementary batteries or other electrical cranking energy shall not be used.

2.8 Starting aids which are permanently mounted on the truck and normally used in starting the engine may be used.

2.9 Smooth engine running without unnatural or continued control manipulation and without use of primers and priming fluids should be obtained within 15 minutes after starting.

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TEST METHOD NO. 12 CONTINUED

2.10 When started, run engine 15 minutes. After completion of the 15 minute operation, operate the engine as follows:

2.10.1 10 minutes idling.

2.10.2 10 minutes half throttle.

2.10.3 5 minutes full throttle.

Perform the 25 minute cycle three times.

2.11 Record temperatures proir to engine starting, immediately after engine starting, and at 10 minute intervals during engine warmup and cycling, including but not limited to the following:

2.11.1 Battery.

2.11.2 Hydraulic tank.

2.11.3 Engine oil.

2.11.4 Engine coolant.

2.11.5 Transmission.

2.12 During engine cycling, and while wearing mittens and boots, the operator shall perform each of the following:

2.12.1 Engage and disengage power train to the wheels in all gears.

2.12.2 Operate through two complete movement cycles all brakes, hoist mechanism, transmission gears, drive unit, safety devices, and horn.

2.12.3 Mount to and dismount from the operator's position on the truck.

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TEST FORM NO. 13

TORQUE CONVERTER

1. Purpose: To determine whether the transmission cooling system and transmission are within specified limits.

2. Test Procedure:

2.1 Install thermocouple in transmission sump.

2.2 Install vacuum gage in engine intake manifold.

2.3 Install tachometer to read engine revolutions per minute.

2.4 With the ambient temperature not less than 30° F (-1$_0$C) , operate the truck until the torque converter oil outlet temperature is 157° F (69$_0$C) or higher. Shift into forward gear and stall the torque converter by applying the service brakes for not less than 60 seconds while the engine is operated at wide open throttle.

2.5 Record torque converter oil temperature, engine vacuum, and engine rpm at 10 second intervals.

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TEST METHOD NO. 14

PERFORMANCE

1. Purpose. To determine the truck can sucessfully complete 1440 laps of the test course without failure. However, 1440 caps must be completed without any failures. If failure(s) occur before completion of 1440 laps, the failure(s) must be corrected and test restarted beginning at the first circuit. Failure for the purpose of this test is defined as any malfunction which cannot be remedied by adjustment, repair, or replacement action by contractor's maintenance personnel using contractor's tools and parts within 30 minutes or the totaled time of the all malfunctions exceeds 5% of the time required to complete 1440 laps and which may cause:

1.1 Failure to commence operation, cessation of operation or degradation of performance below designated levels.

1.2 Serious damage to truck by continued operation.

1.3 Serious personnel safety hazard. Only routine maintenance as prescribed by the manufacturer is allowed. The truck shall show no signs of unusal or detrimental wear. At the completion of the test, the truck shall be further tested to show it is capable of meeting all the performance criteria set forth herein.

2. Performance test.

2.1 Layout of course.

2.1.1 Description. The test course shall be set up in accordance with the provisions of Figure 1. Course length shall be as specified except the following deviations will be permitted: The course should be laid out to include natural grades of the slope and approximate length specified for the ramp. Only if it is impossible to do this should a ramp be constructed.

2.1.1.1 Aisle or road widths. Aisle or road widths shall be 20 feet (6096 mm). The entire perimeter of the course shall be encompassed with a suitable material to form a barrier having a height of at least 12 inches (305 mm) . Rubber pylons, or other suitable marking devices, shall be placed at all corners and other critical points to assist in keeping the machine under test within the course aisle.

2.1.2 Course illustrated by Figure 1.

2.1.3 Specification of course features.

2.1.3.1 Obstacle course. The obstacle course shall be set up in accordance with the provisions of Figure 2, as a part of the test course. A guide line equal in length to two times the wheelbase of the truck under test shall be provided parallel and to the left of the center line of the course. This shall be at a distance from the course center line equal to 1/2 the distance between front wheel centers. The truck shall traverse the obstacle course in a straight line with its left front wheels tracking on the guide line. The spacing and arrangement shall allow the right front wheel tracking on the guide line. The spacing and arrangement shall allow the right front and left rear wheels to be on the crest of blocks simultaneouly while the other two

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TEST METHOD NO. 14 CONTINUED

wheels are at ground level. Similarly, the left front and right rear wheels shall be elevated simultaneously while the other two wheels are at ground level. This rocking action shall occur while the machine is moving forward over the obstacle course.

2.1.3.2 Specification of standard test loads. Test loads used in the performance test shall be in accordance with 4.3.2(c).

2.1.3.3 Specification of load areas. Skidded test loads shall be positioned in accordance with Figure 1.

2.1.3.4 Minimum lighting requirement. Headlights of the machine under test shall be on at all times during the performance test.

2.1.3.4.1 Light switch. The light switch shall be turned off and on at the beginning of each lap.

2.1.3.5 Course surface. Except where otherwise specified, the test course shall be covered with hard-packed earth or gravel, with a surface permitting tire impressions not greater than 1/4 inch (6 mm) deep when the carrier is transporting a capacity load, or any other surface or paving which meets this requirement.

2.2 Standard procedures.

2.2.1 Speed of negotiating course.

2.2.1.1 General. Truck(s) undergoing test should traverse each part of the performance test course at the maximum safe practical speed.

2.2.1.2 Laps per hour. Truck(s) shall complete 12 ciruits of the test course during each individual operating hour of the performance test.

2.2.1.3 Laps per 8-hour day. Truck(s) shall complete 96 circuits of the test course during each individual operating 8-hour day of the performance test without requiring more than 1 refueling.

2.2.1.4 Duration of the performance test. The performance test shall be conducted 8 hour per day, 5 days per week, for a period of 15 operating 8-hour days of test. By means of the accelerated wear features built into the performance test, this period is intended to be the equivalent of approximately 6 months of normal truck operation.

2.2.2 Conditions of course during test. The test course shall be free of snow, ice, and non-planned obstacles, or foreign material during the course of the test.

2.2.3 Maintenance service. Maintenance service shall be given according to the provisions of the "Maintenance Service Sheet" in accordance with Figure (3). This procedure shall be carried out prior to the endurance testing. In addition, to the initial maintenance service, which includes the operator's daily responsibility, such other repairs shall be in accordance with 1.1.

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TEST METHOD NO. 14 CONTINUED

2.2.4 Method of negotiating course. Truck(s) under test shall begin each lap at the test load area marked (A) on figure 1 and in a position to pick up the test load. The operator shall then sound the horn for approximately 1 second, elevate the test standard load to the travel position and perform the following sequence of operations to complete the test course:

2.2.4.1 Back into the transverse leg of the test course without backing into the portion of course marked area (B).

2.2.4.2 Drive forward making a 90° left hand turn and proceed along area (B).

2.2.4.3 At intersection, make 90° right hand turn until adjacent to area marked (C).

2.2.4.4 Ground the test load and back out into main aisle.

2.2.4.5 Return to the straddle position over the load and elevate the test load to the carry position.

2.2.4.6 Back out into the aisle area (B) from which position make a 90° left hand turn and proceed along area marked (D).

2.2.4.7 At intersection, make a 90° left hand turn onto the ramp without backing.

2.2.4.8 Stop truck when arriving at position on grade labeled "Stopping Point".

2.2.4.9 Shift gears, if necessary, and resume mounting grade or ramp.

2.2.4.10 Proceed down grade making a right hand turn off of the ramp and left hand turn into portion of course marked (F).

2.2.4.11 Make left hand turn into area (G) and proceed to obstacle course area.

2.2.4.12 Traverse obstacle course in accordance with 2.1.3.1. This part of the test may be conducted at reduced speed with the transmission in low gear, however, the truck shall not be stopped while going over the blocks.

2.2.4.13 Upon completion of the obstacle course, the truck shall be driven to the initial starting area (A). The test load shall then be grounded and the lifting mechanism fully lowered. The driver shall continue to complete additional circuits of the test course in accordance with 2.2 through 2.2.4.

2.2.4.14 On alternate days, the course shall be traversed in the opposite (or clockwise) direction. On these days, the test procedure shall be reversed.

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TEST METHOD NO. 14 CONTINUED

2.2.4.15 The "Hourly Time Record Sheet" (see figure (5)) shall be filled out for each operating hour of the performance test.

2.2.5 Operating limitations. The truck under test shall not be permitted to leave the bounds circumscribed in Figure 1, nor shall the procedure requirements specified herein be altered.

3. Post-test.

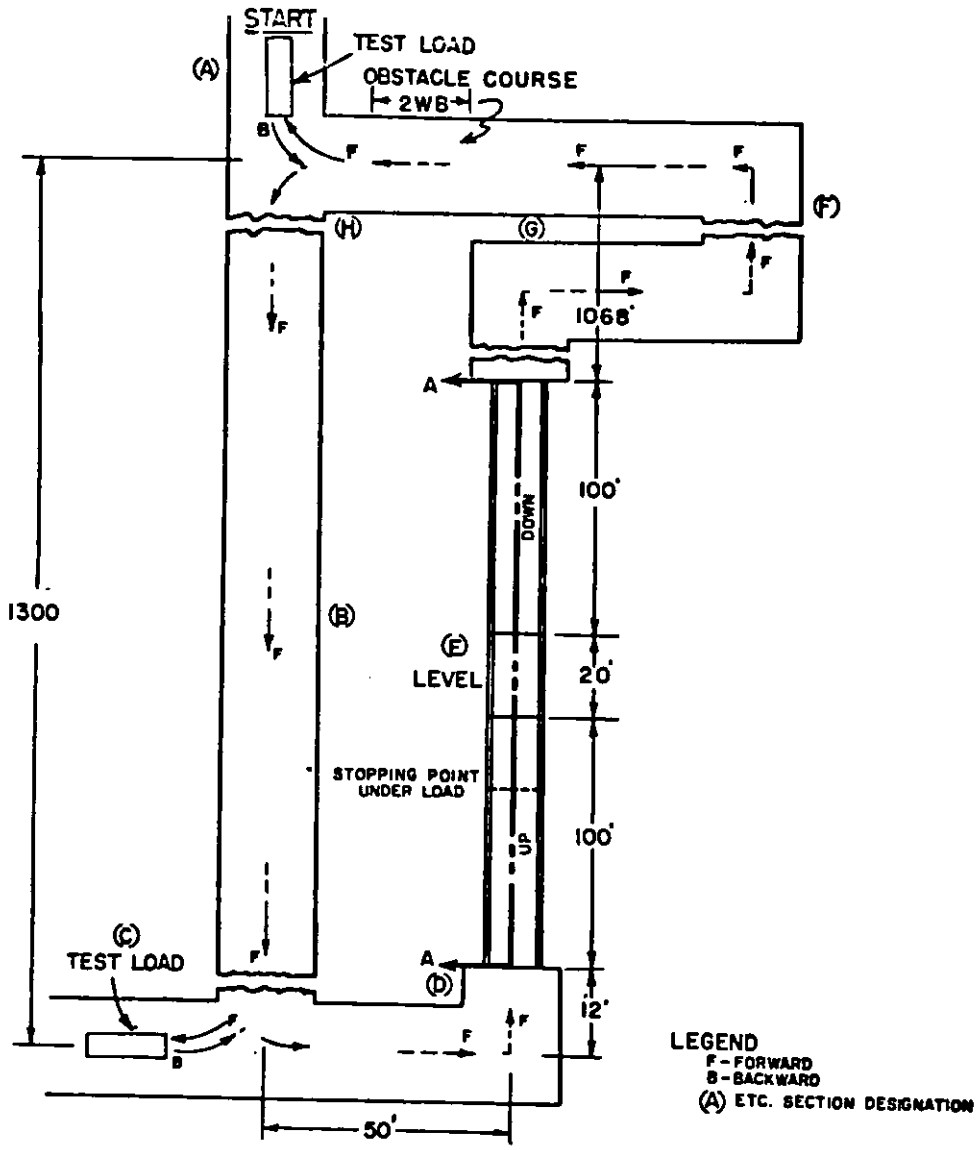
3.1 The post-test inspection shall be conducted after the truck(s) has completed the performance test or if it fails the performance test, the inspection portion of this procedure shall be conducted after such a failure.

3.2 Composition. the post-test consists of two parts:

3.2.1 Post-Test Inspections (see Table III)

3.2.2 Post-Tests (see Table IV)

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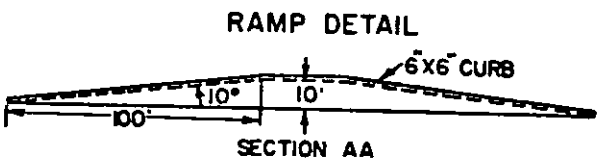
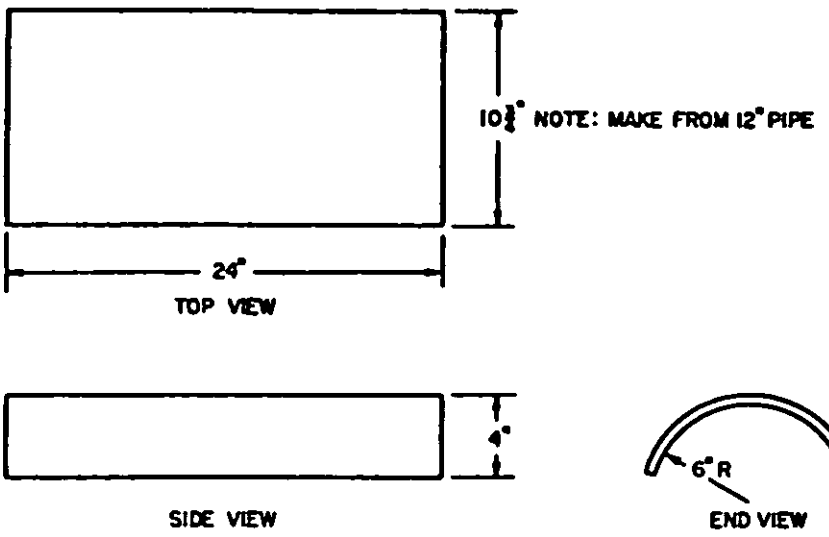
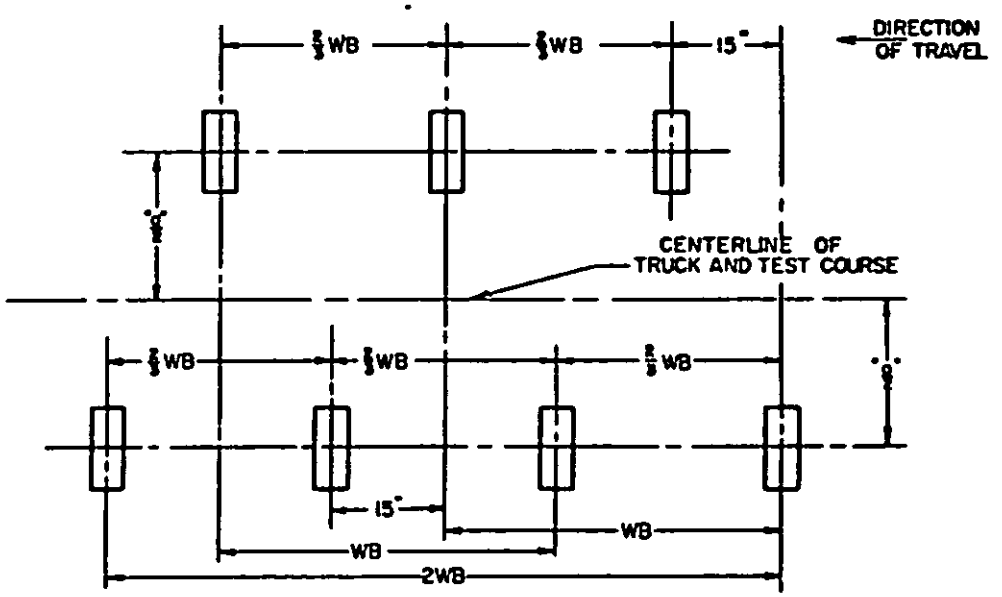


FIG. 1 – OUTDOOR TEST COURSE FOR TRUCK, STRADDLE-CARRY, GASOLINE

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LAYOUT OF OBSTACLE COURSE



NOTE:
"a" = DISTANCE BETWEEN CENTERLINES OF DRIVING WHEELS
WB = WHEEL BASE

AREA "a" ON TEST COURSE

FIG.2 DETAILS OF OBSTACLE CONSTRUCTION

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MAINTENANCE SERVICE SHEET

1. Equipment shall be serviced and inspected in accordance with the following maintenance schedule proir to and during performance testing to insure maximum operational readiness. Any excessive usage of lubricants or fluids or excessive wear, or excessive adjustment shall be listed on the performance test report and reported to the contracting officer.

Every 8 Hours or Daily

(a) Check engine lubricant and cooling system levels
(b) Check and fill if necessary tire wear and pressure
(c) Check battery liquid level and fill if necessary
(d) Check transmission fluid level and fill if necessary
(e) Check power steering fluid level and fill if necessary
(f) Check hydraulic system fluid level and fill if necessary
(g) Lubricate equipment in accordance with manufacturer's recommendations.
(h) Check the brake fluid level and fill if necessary

At 60 Hours

(a) Clean or replace all engine, hydraulic, and transmission filters
(b) Replace fuel filter(s)
(c) Clean the breather for the hydraulic system
(d) Check and adjust all drive belts
(e) Check and adjust the parking brake for adjustment
(f) Check and adjust the service brakes
(g) Check, clean, or replace the air filter if necessary

At 120 Hours

(a) Repeat all operations for 60 hour maintenance check
(b) Change engine oil, hydrauic, and fuel filters
(c) Change air filter
(d) Check hydraulic and transmission drive shafts
(e) Check steering tie rods
(f) Check axles and differential
(g) Check wheel bearings
(h) Check brake shoe wear

Figure 3

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TABLE V - MILITARY LUBRICANTS

ITEM APPLICATION	SPECIFICATION	TYPE & GRADE
1. Engine Oils		
a. Heavy duty	MIL-L-2104	10, 30
b. Operational and preservative (preservation only)	MIL-L-21260	Grade 1, 2, & 3
c. Arctic	MIL-L-46167	One Grade
2. Fuel		
Diesel	VV-F-800	DF-1, DF-2, & DF-A
3. Grease		
Automotive	MIL-G-10924	One Grade
4. Hydraulic-system oil and transmission fluid		
a. Oil	MIL-L-2104	10
b. Arctic (for consistent operation (below -10 F)	MIL-L-46167	One Grade
5. Lubricating oils		
a. Chain, gear, wire rope	VV-L-751	A, B, C
b. Universal	MIL-L-2105	75, 80, 90
c. Arctic	MIL-L-10324	One Grade
6. Anti-Freeze		
a. Ethylene glycol	MIL-A-46153	
7. Brake Fluid	VV-B-680	One Grade

Figure 4

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HOURLY TIME RECORD SHEET

TRUCKS, STRADDLE-CARRY, DIESEL,
PNEUMATIC RUBBER TIRES

Truck Manufacturer	Model	Serial No.	Date

Test Location	Recorded By:	Clock Readings: (start)	(finish)	Required laps/hr

Laps Number	Watch Reading	Elasped Time	Remarks
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
Total			Average Lap Time_____

Instructions for Timing Performance Test

1. Start decimal minute stop watch at zero at beginning of each operating hour.
2. Record watch reading at end of each lap in "Watch Reading" space.
3. Subtract watch reading at end of preceding lap from reading at end of lap. Record differences in "Elapsed Time" space.
4. If equipment under test completes the required number of laps prior to completion of an operating hour, is shall be stopped until start of next hour, and then resume performance test.

NOTE: This form is for use in conjunction with PERFORMANCE TEST METHOD NO. 14.

Figure 5

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TEST METHOD NO. 15

DRIFT

1. Purpose. To determine if the vertical drift is within specified limits.

2. Test Course. Level, flat surface.

3. Test Apparatus. Temperature thermocouples and clinometer.

4. Test Procedure:

   4.1 Install thermocouple in the hydraulic reservior. Heat the hydraulic oil by exercising the lift function and maintain hydraulic oil temperature above specified temperature during the duration of the test.

   4.2 Raise rated load to maximum lift height with shoe lift rods in true vertical position measured with the clinometer. Mark reference point on wheel guard.

   4.3 Record hydraulic oil temperature.

   4.4 Hold for specified time and record downward drift.

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TEST METHOD NO. 16

VISIBILITY

1. Purpose. To determine that the visibility from the operator's cab complies with specified requirements.

2. Test Course. Level, flat surface.

3. Test Procedure:

3.1 Place truck on dry, level surface with operator in cab.

3.2 Use a four foot high pole with stand, place directly in front of truck, at truck midpoint, and up against the truck frame. Determine if the operator can see the pole by using mirrors or by direct sight.

3.3 Place the pole in front of the right front and then left front wheel and record visibility. Place pole at three separate locations along each side of the truck and at equal distances apart starting from the front of the truck and record visibility. Place the pole in front of the left rear, right rear, and then directly behind truck at truck midpoint and record visibility.

3.4 Repeat the procedure in paragraph 3.2 and 3.3 moving the pole out from the truck one foot at each position described. Movement out from the tires shall be at 45 degree angle away from the truck. Repeat process until final location of pole is 40 feet away from truck.

3.5 Record visibility at all locations indicated whether sight is through mirrors or direct sight.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-16101D	TRUCKS, STRADDLE-CARRY, DIESEL, 30,000 POUND CAPACITY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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